The programmer supports reading and writing flash and EEPROM chips used in display boards without removing them, using the dedicated VGA or HDMI sockets.

It also includes **infrared (IR) transmit and receive functionality**, allowing remote control code communication and modification.

Practical Implementation of the Advanced Programming Tool

A practical implementation of the highly efficient programming tool that is essential for maintenance technicians.

POSTAL3 - AVR - USB

It supports:

25CXX – 93CXX – 24CXX 45DXX – 25EXX – EMMC – JTAG – I2C – VGA + HD

The programmer operates via USB only.

It delivers high-speed performance using the latest:

CP210x Bridge USB XPRESS - X2

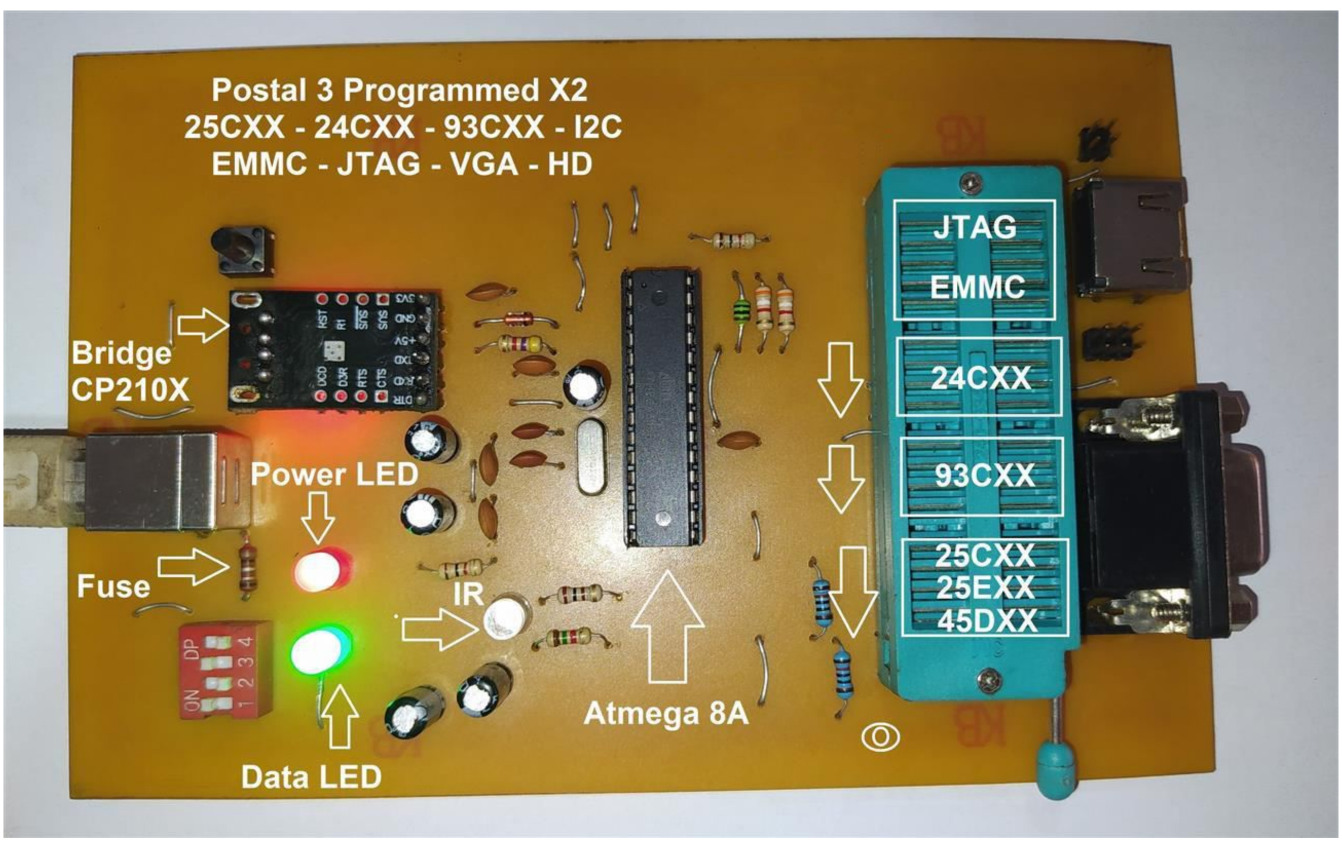
The programmer is designed for stable operation with no errors.

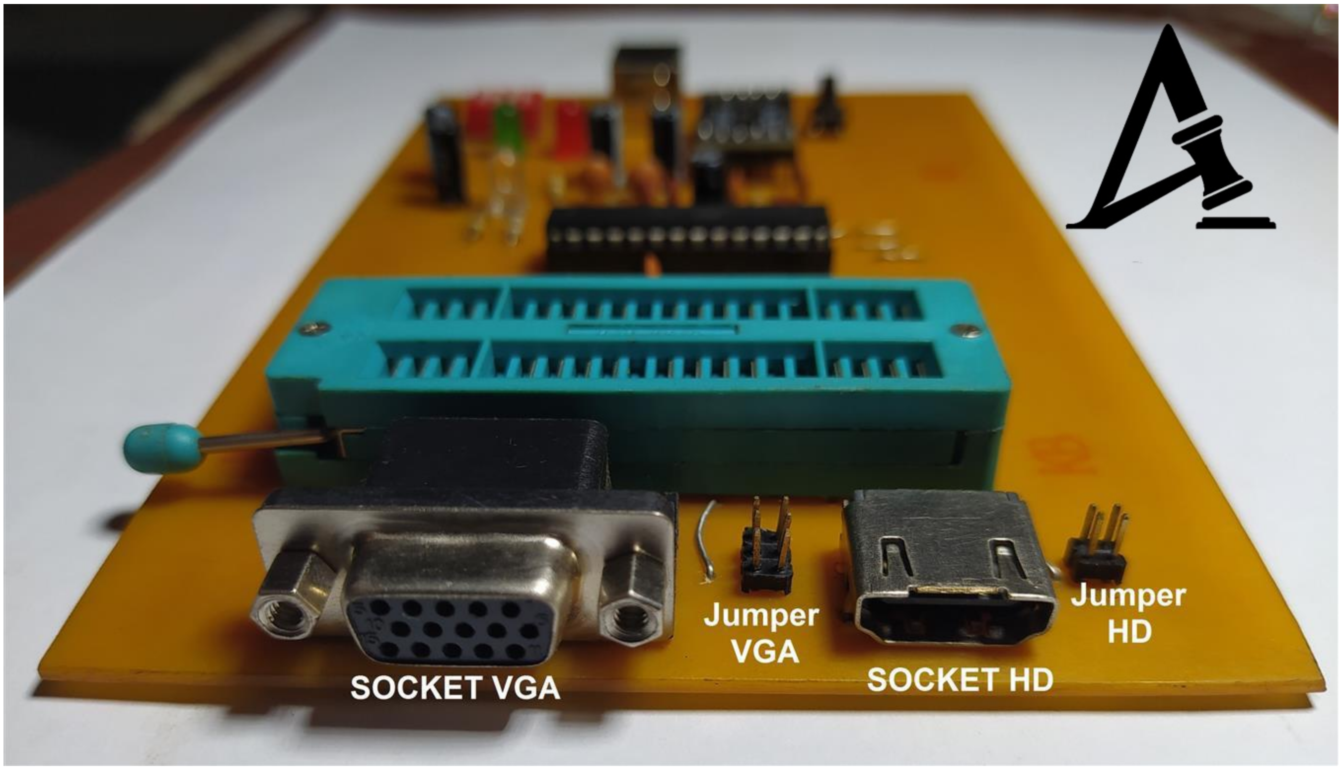
All components used are original and high quality.

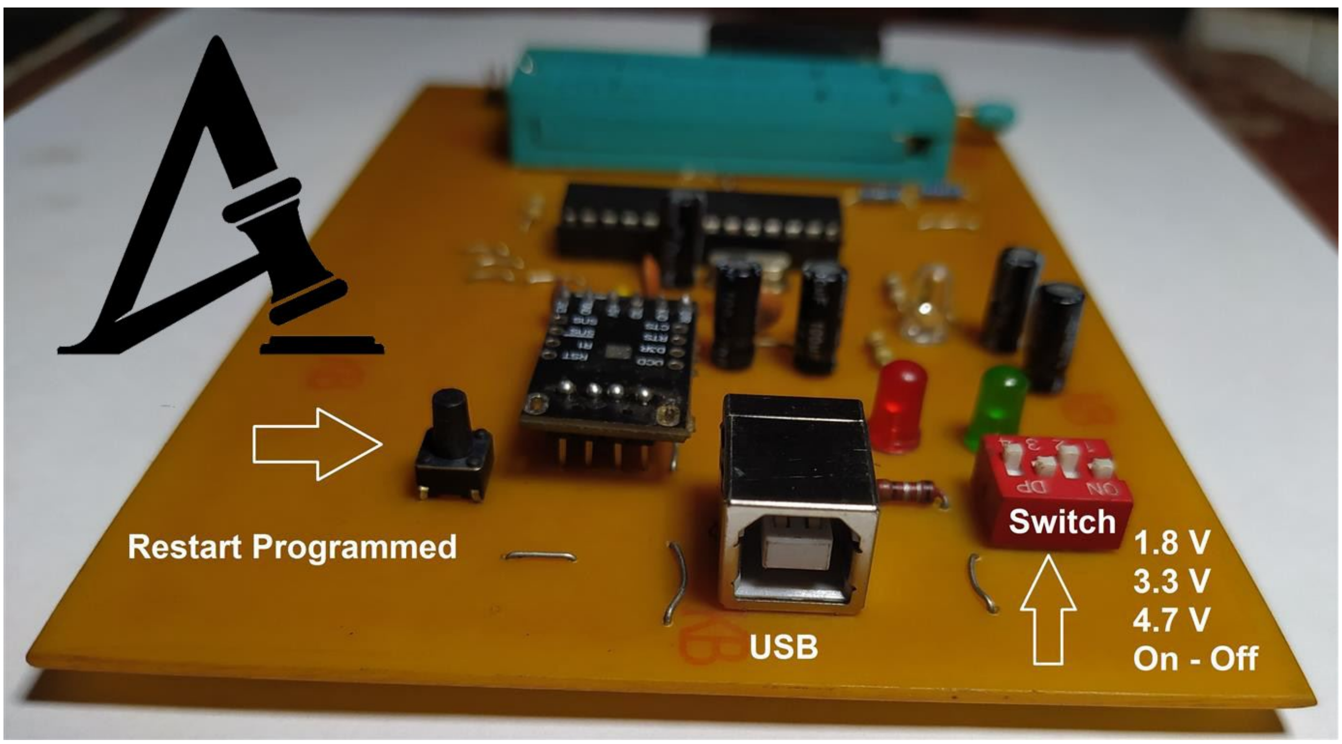
The programmer has been fully tested before release and has demonstrated excellent performance and reliability

All components and power measurements are within normal and stable ranges.

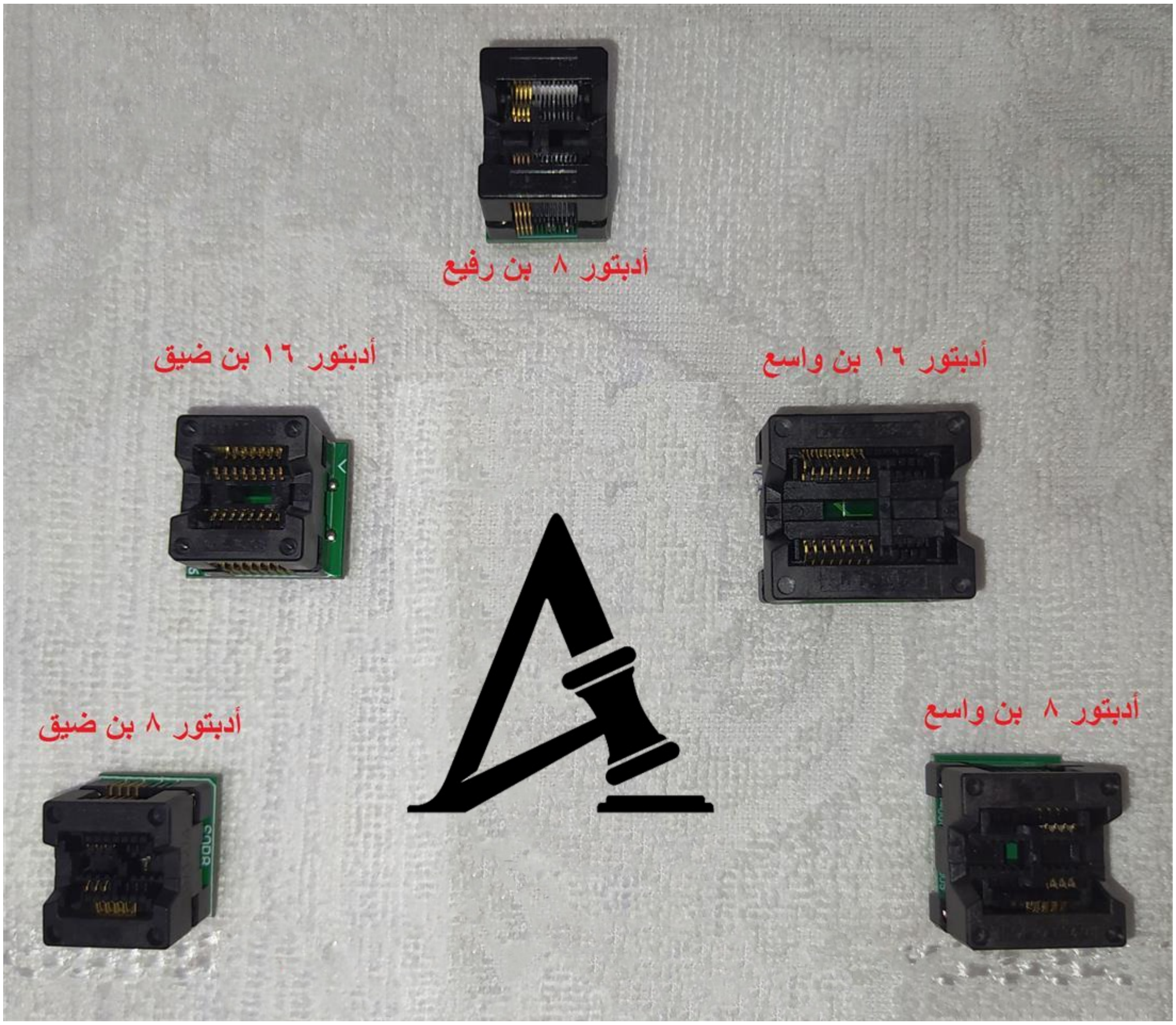
Programmer Images





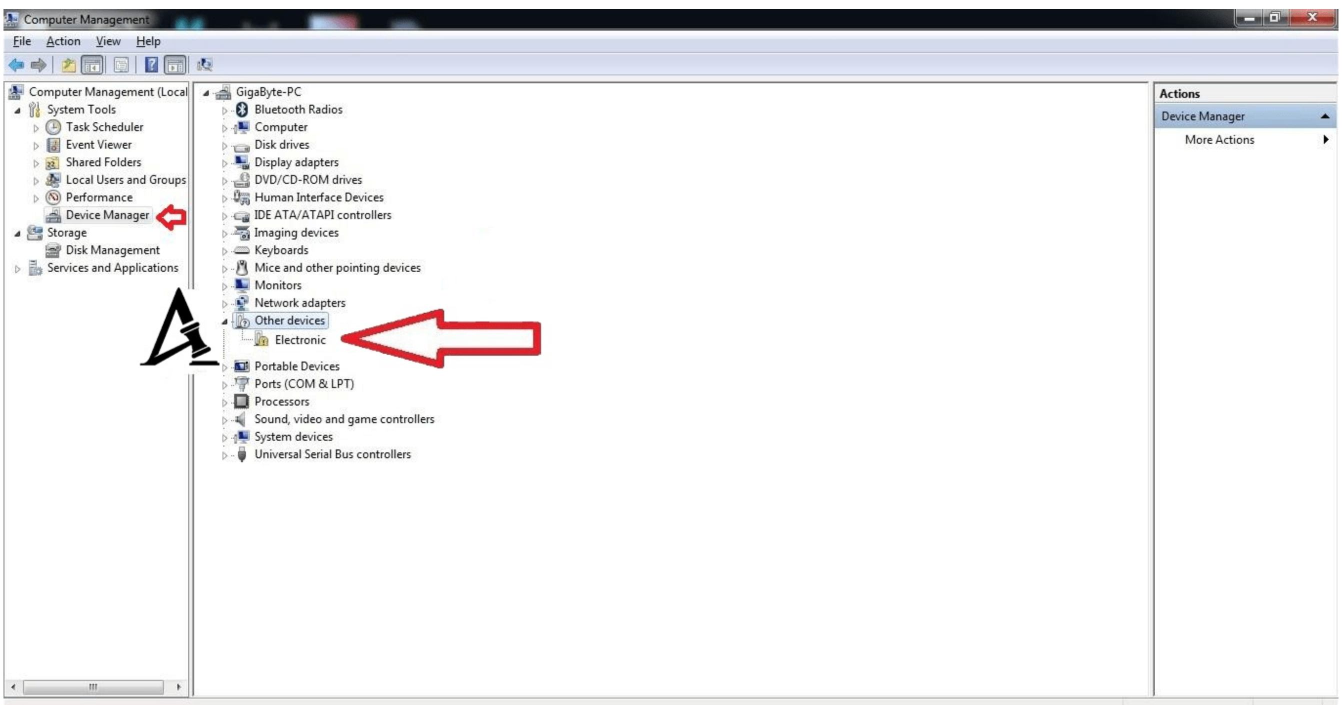


Programmer Adapters

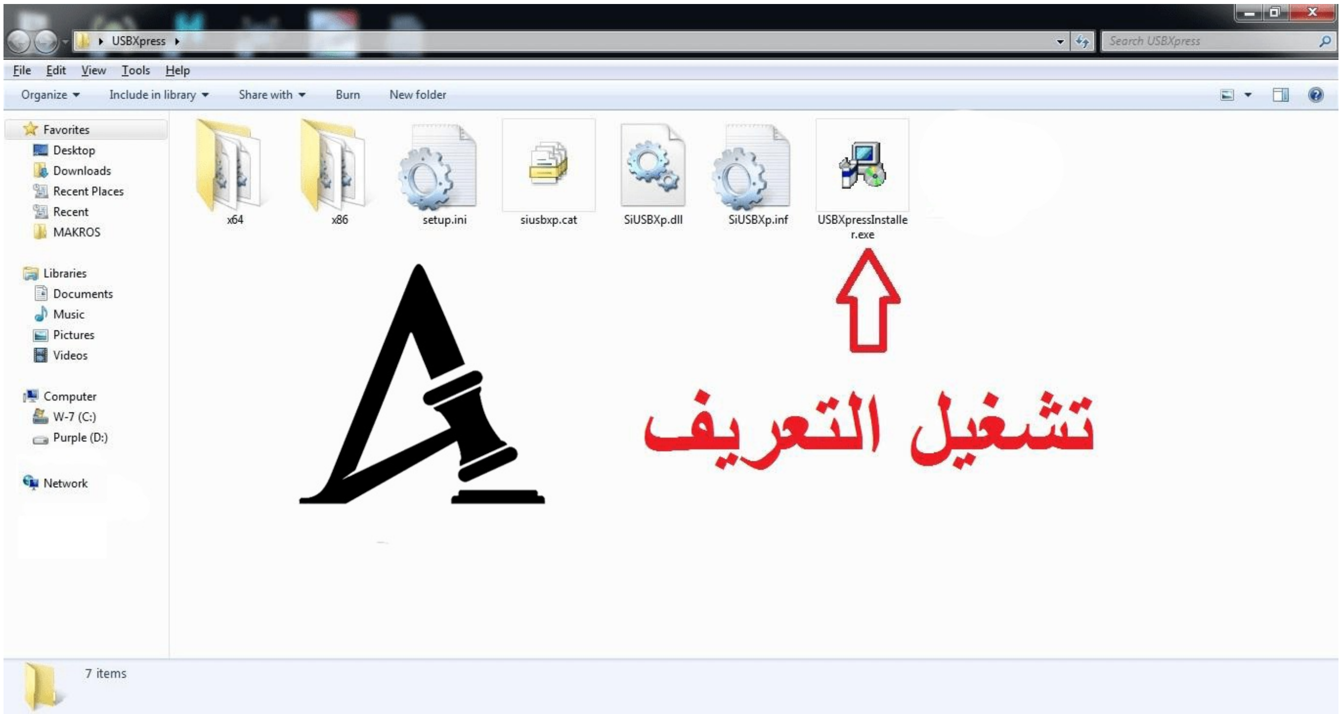


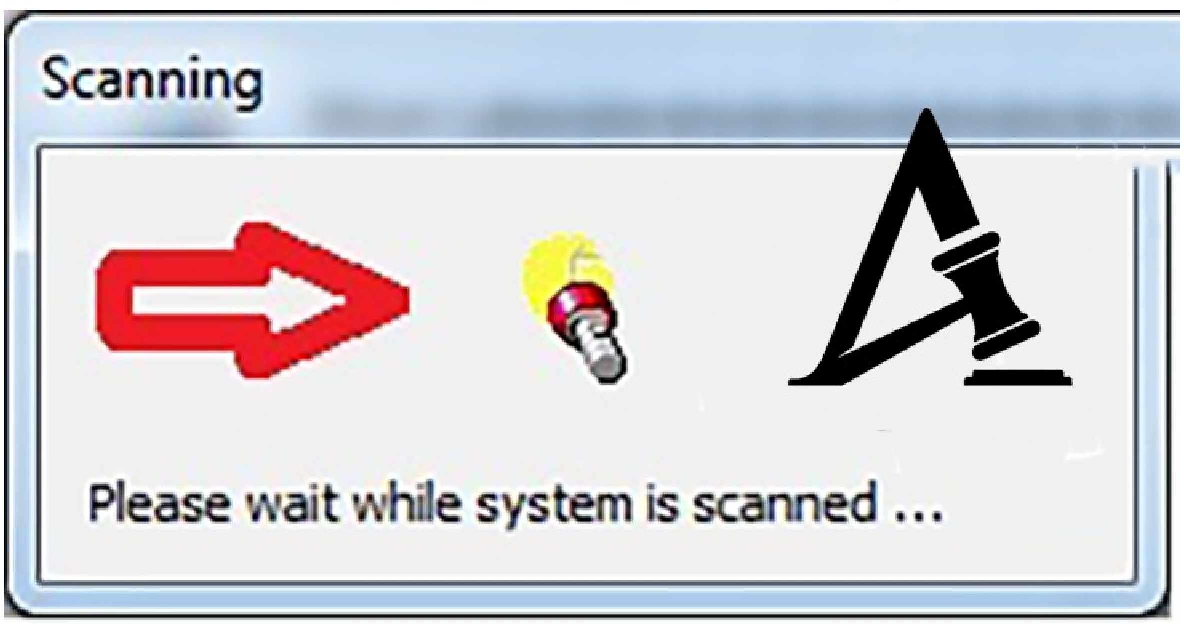
Driver Installation Steps for the Programmer

When connecting the programmer to the USB port for the first time only.

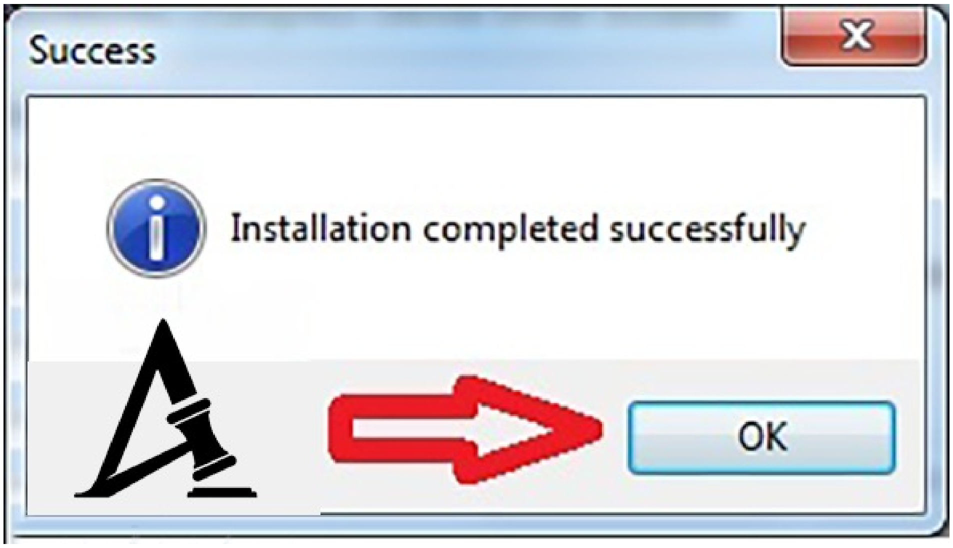


Open the driver folder

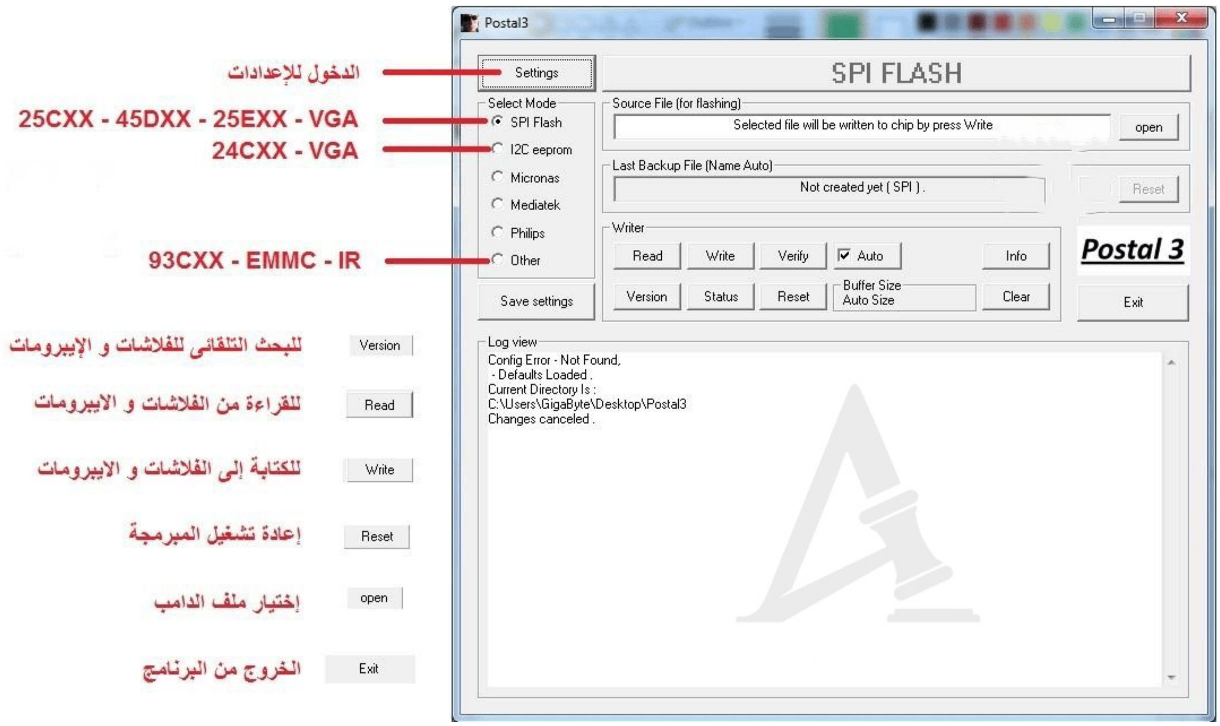




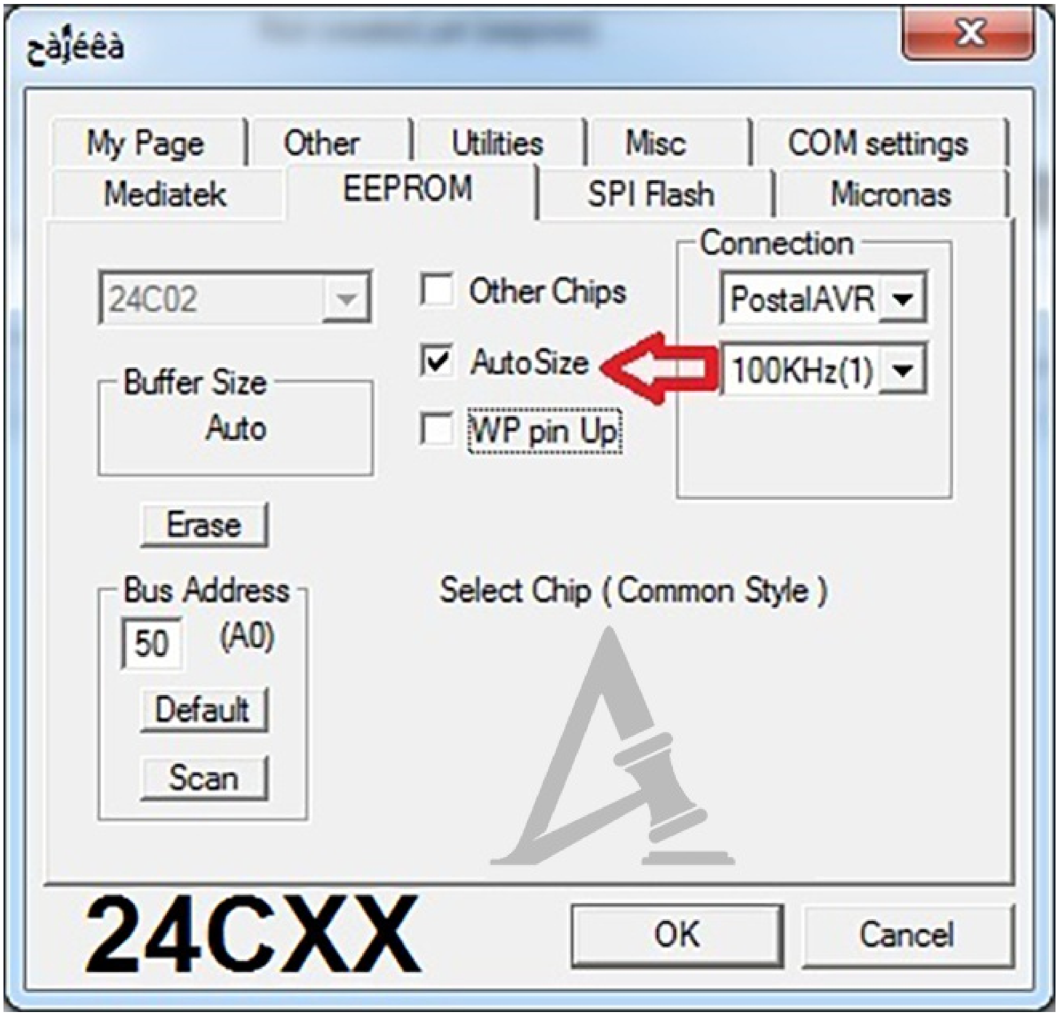
> The device has been successfully recognized on Windows.

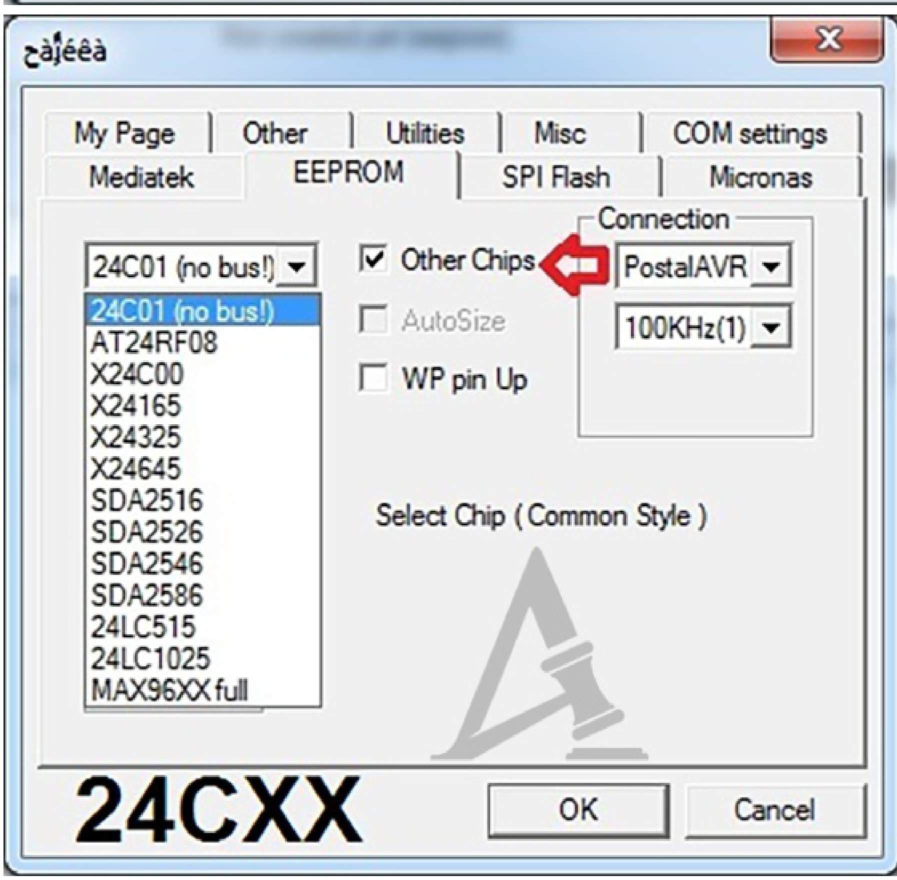


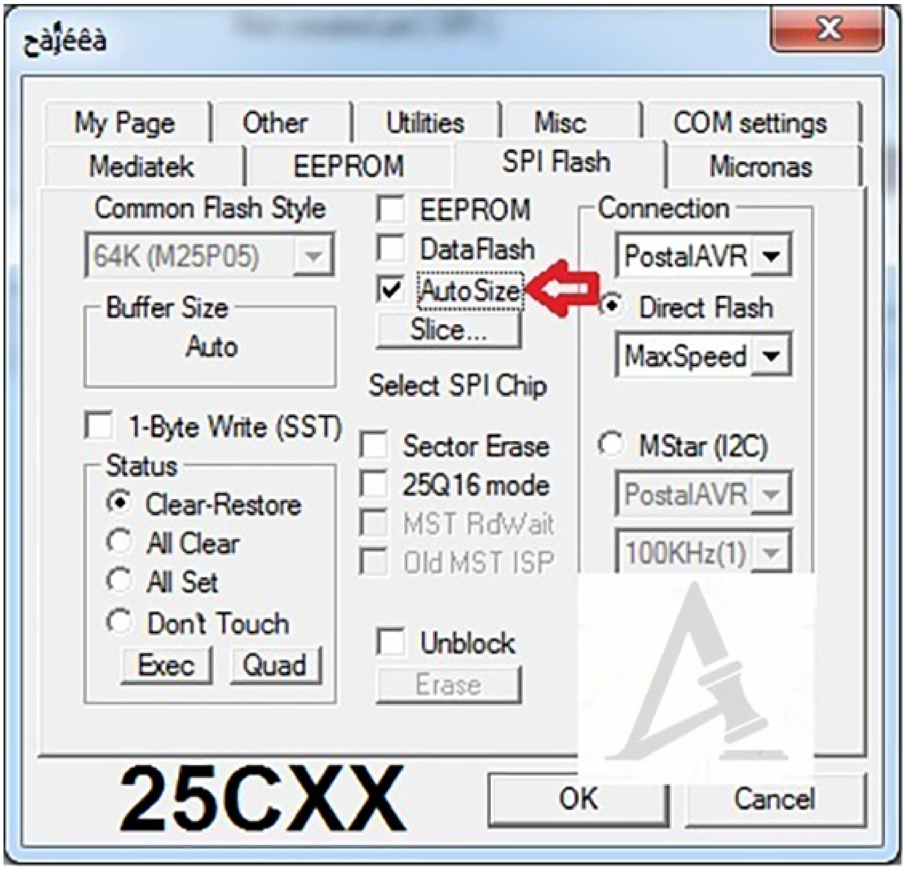
Overview of the POSTAL3 Software Interface

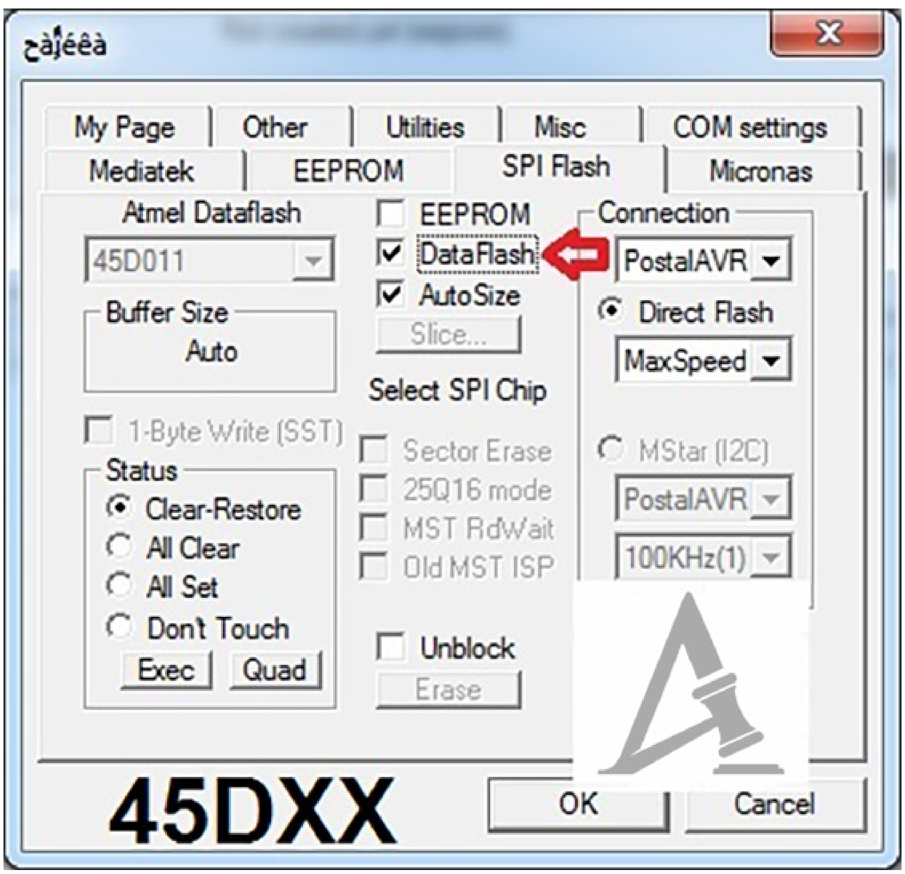


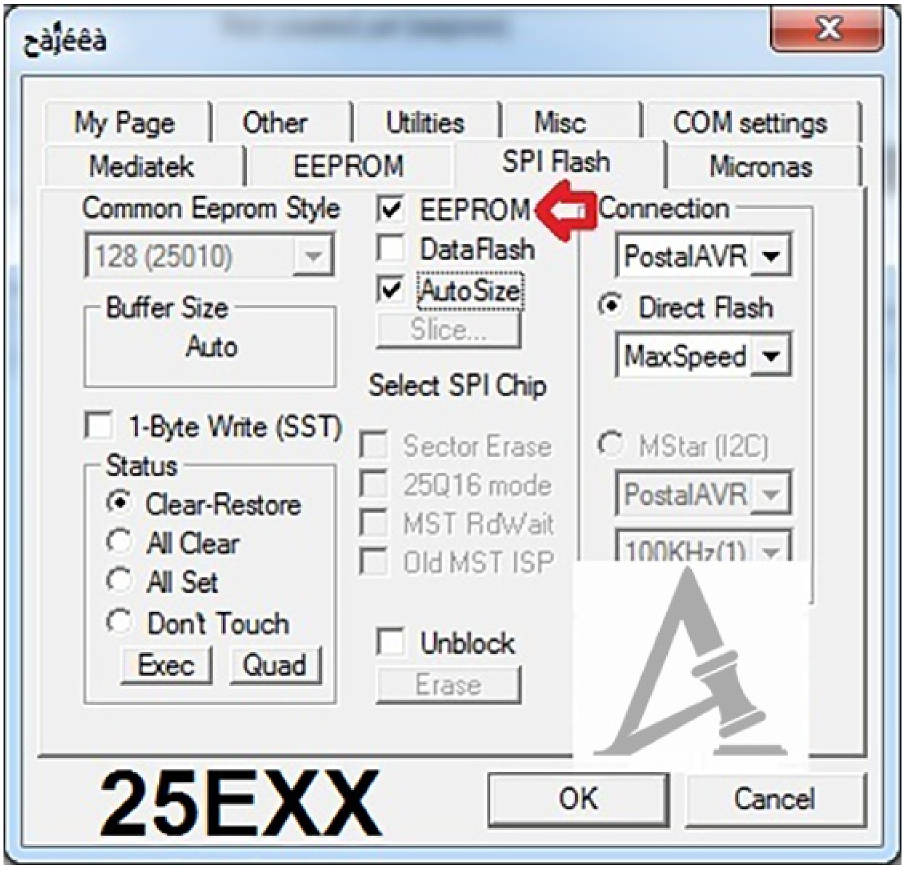
Basic Settings of the POSTAL3 Software

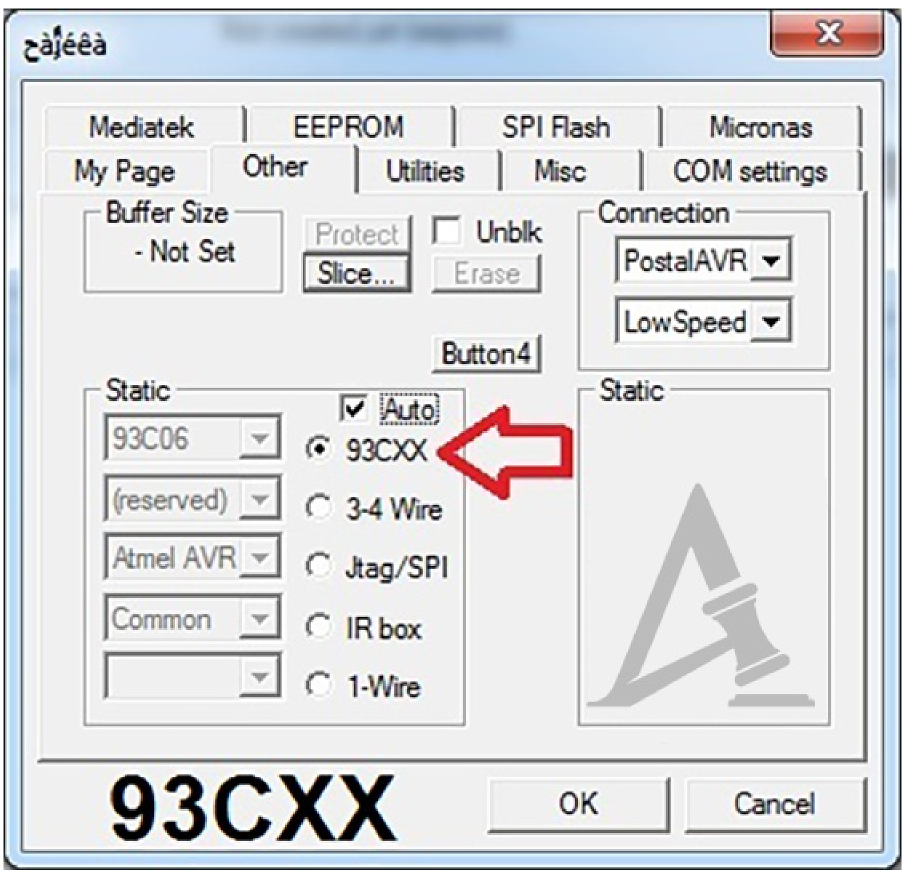

My Page
Other
Utilities
Misc
COM settings
Mediatek
EEPROM
SPI Flash
Micronas
24C02
Other Chips
AutoSize
WP pin Up
Connection
PostalAVR
100KHz(1)
Buffer Size
Auto
Erase
Bus Address
50
(A0)
Default
Scan
Select Chip (Common Style)
24CXX
OK
Cancel

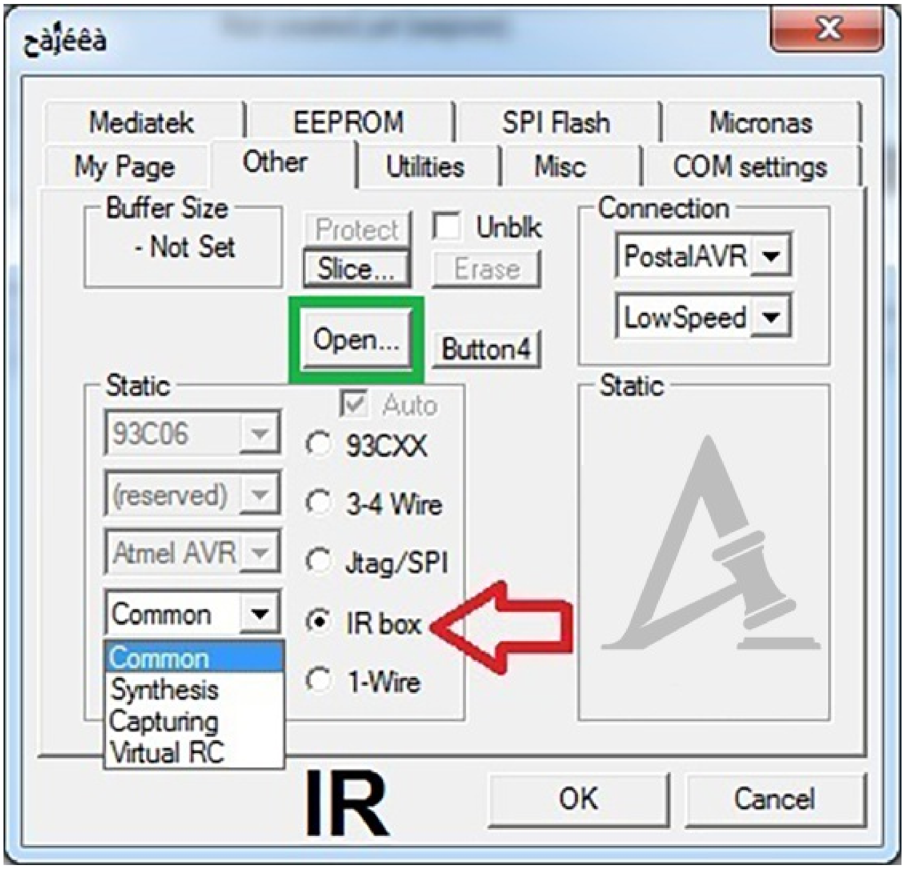

My Page
Other
Utilities
Misc
COM settings
Mediatek
EEPROM
SPI Flash
Micronas
24C01 (no bus!)
24C01 (no bus!)
AT24RF08
X24C00
X24165
X24325
X24645
SDA2516
SDA2526
SDA2546
SDA2586
24LC515
24LC1025
MAX96XX full
Other Chips
AutoSize
WP pin Up
Connection
PostalAVR
100KHz(1)
Select Chip (Common Style)
24CXX
OK
Cancel

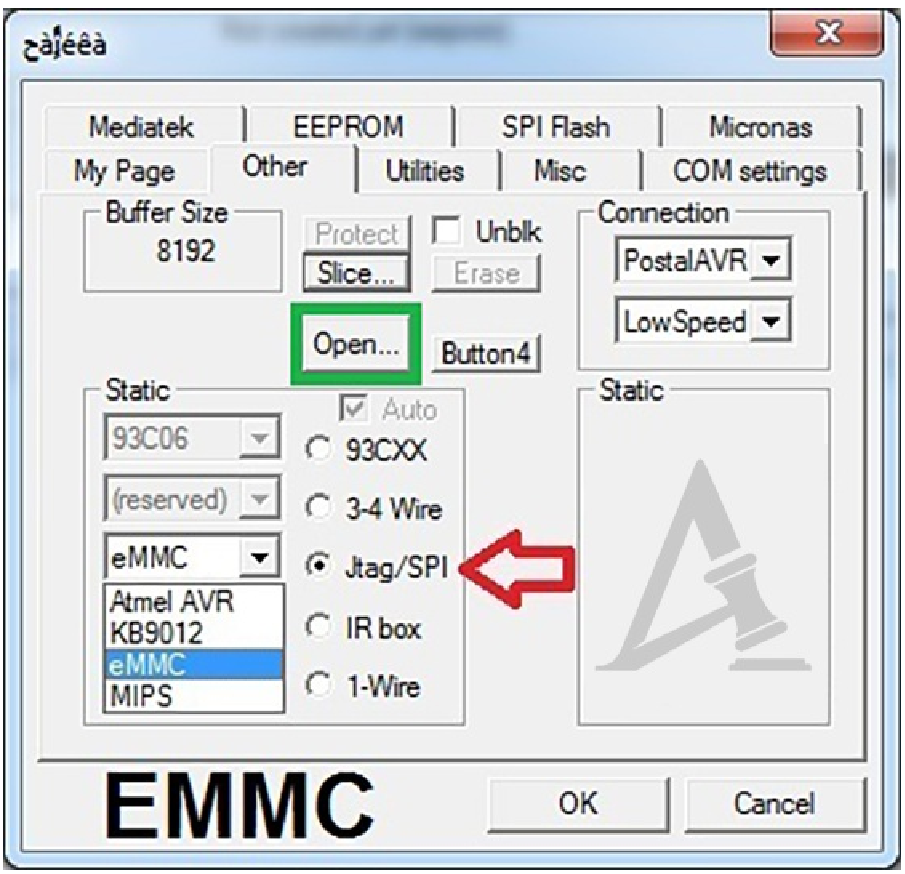

My Page
Other
Utilities
Misc
COM settings
Mediatek
EEPROM
SPI Flash
Micronas
Common Flash Style
64K (M25P05)
Buffer Size
Auto
1-Byte Write (SST)
Status
Clear-Restore
All Clear
All Set
Don't Touch
Exec
Quad
EEPROM
DataFlash
AutoSize
Slice...
Select SPI Chip
Sector Erase
25Q16 mode
MST RdWait
Old MST ISP
Unblock
Erase
Connection
PostalAVR
Direct Flash
MaxSpeed
MStar (I2C)
PostalAVR
100KHz(1)
25CXX
OK
Cancel

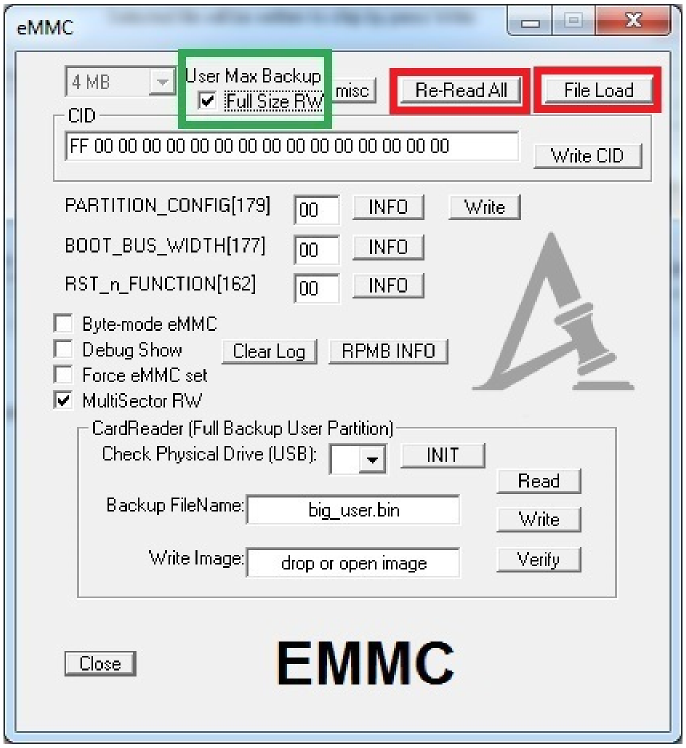

My Page
Other
Utilities
Misc
COM settings
Mediatek
EEPROM
SPI Flash
Micronas
Atmel Dataflash
45D011
Buffer Size
Auto
1-Byte Write (SST)
Status
Clear-Restore
All Clear
All Set
Don't Touch
Exec
Quad
EEPROM
DataFlash
AutoSize
Slice...
Select SPI Chip
Sector Erase
25Q16 mode
MST RdWait
Old MST ISP
Unblock
Erase
Connection
PostalAVR
Direct Flash
MaxSpeed
MStar (I2C)
PostalAVR
100KHz(1)
45DXX
OK
Cancel

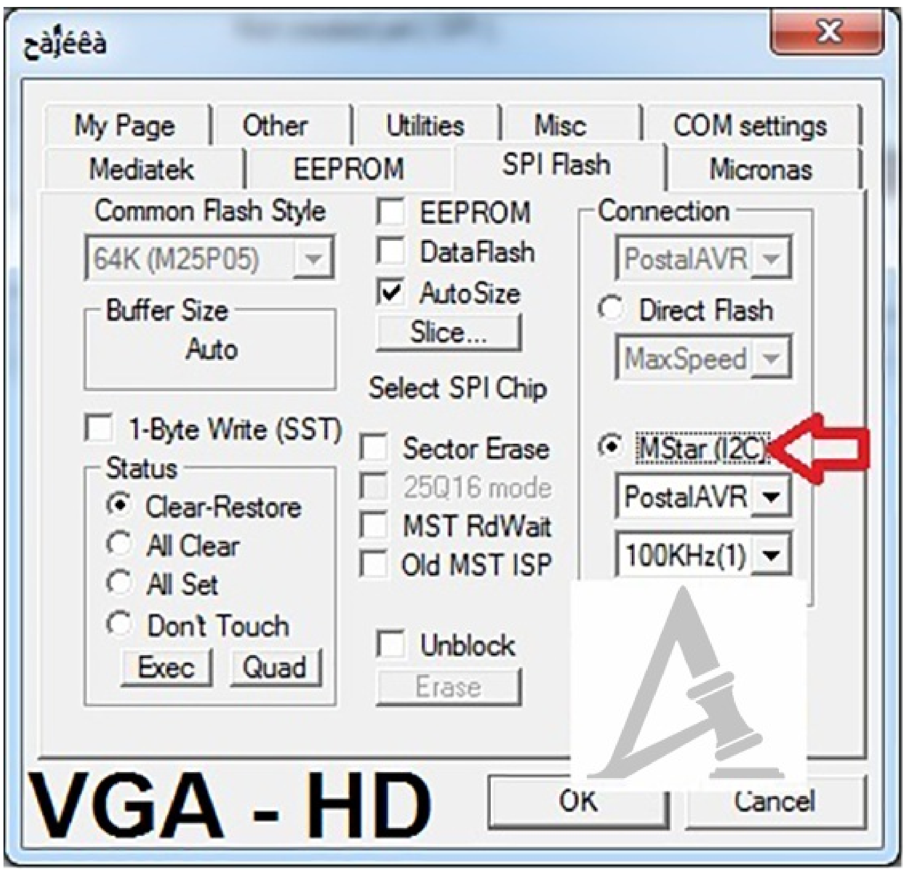

My Page
Other
Utilities
Misc
COM settings
Mediatek
EEPROM
SPI Flash
Micronas
Common Eeprom Style
128 (25010)
EEPROM
DataFlash
AutoSize
Slice...
Connection
PostalAVR
Direct Flash
MaxSpeed
Buffer Size
Auto
Select SPI Chip
1-Byte Write (SST)
Sector Erase
25Q16 mode
MST RdWait
Old MST ISP
MStar (I2C)
PostalAVR
100KHz(1)
Status
Clear-Restore
All Clear
All Set
Don't Touch
Exec
Quad
Unblock
Erase
25EXX
OK
Cancel


Mediatek
EEPROM
SPI Flash
Micronas
My Page
Other
Utilities
Misc
COM settings
Buffer Size
- Not Set
Protect
Slice...
Unblk
Erase
Connection
PostalAVR
LowSpeed
Button4
Static
Auto
93C06
93CXX
(reserved)
3-4 Wire
Atmel AVR
Jtag/SPI
Common
IR box
1-Wire
Static
93CXX
OK
Cancel

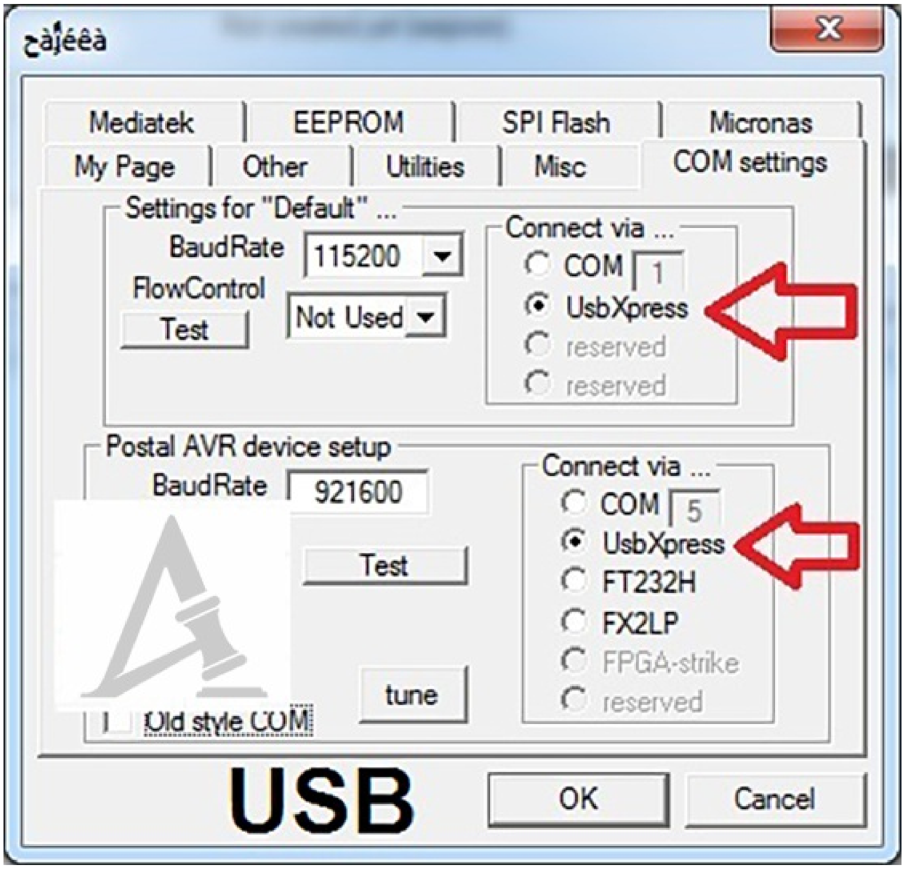

Mediatek
EEPROM
SPI Flash
Micronas
My Page
Other
Utilities
Misc
COM settings
Buffer Size
- Not Set
Protect
Unblk
Slice...
Erase
Connection
PostalAVR
LowSpeed
Open...
Button4
Static
Auto
93C06
93CXX
(reserved)
3-4 Wire
Atmel AVR
Jtag/SPI
Common
IR box
Common
Synthesis
Capturing
Virtual RC
1-Wire
Static
IR
OK
Cancel


Mediatek
EEPROM
SPI Flash
Micronas
My Page
Other
Utilities
Misc
COM settings
Buffer Size
8192
Protect
Unblk
Slice...
Erase
Connection
PostalAVR
LowSpeed
Open...
Button4
Static
Auto
93C06
93CXX
(reserved)
3-4 Wire
eMMC
Jtag/SPI
Atmel AVR
KB9012
IR box
eMMC
MIPS
1-Wire
Static
EMMC
OK
Cancel



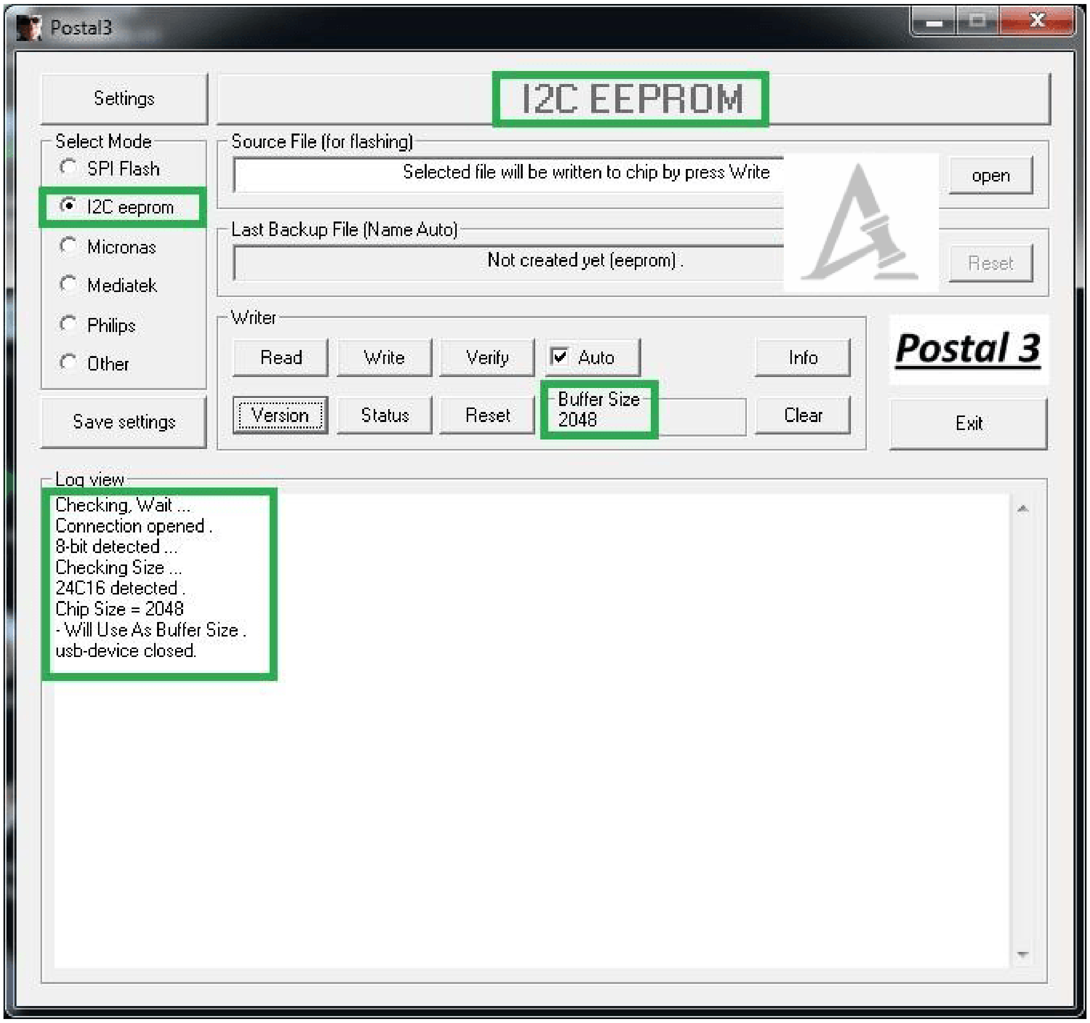


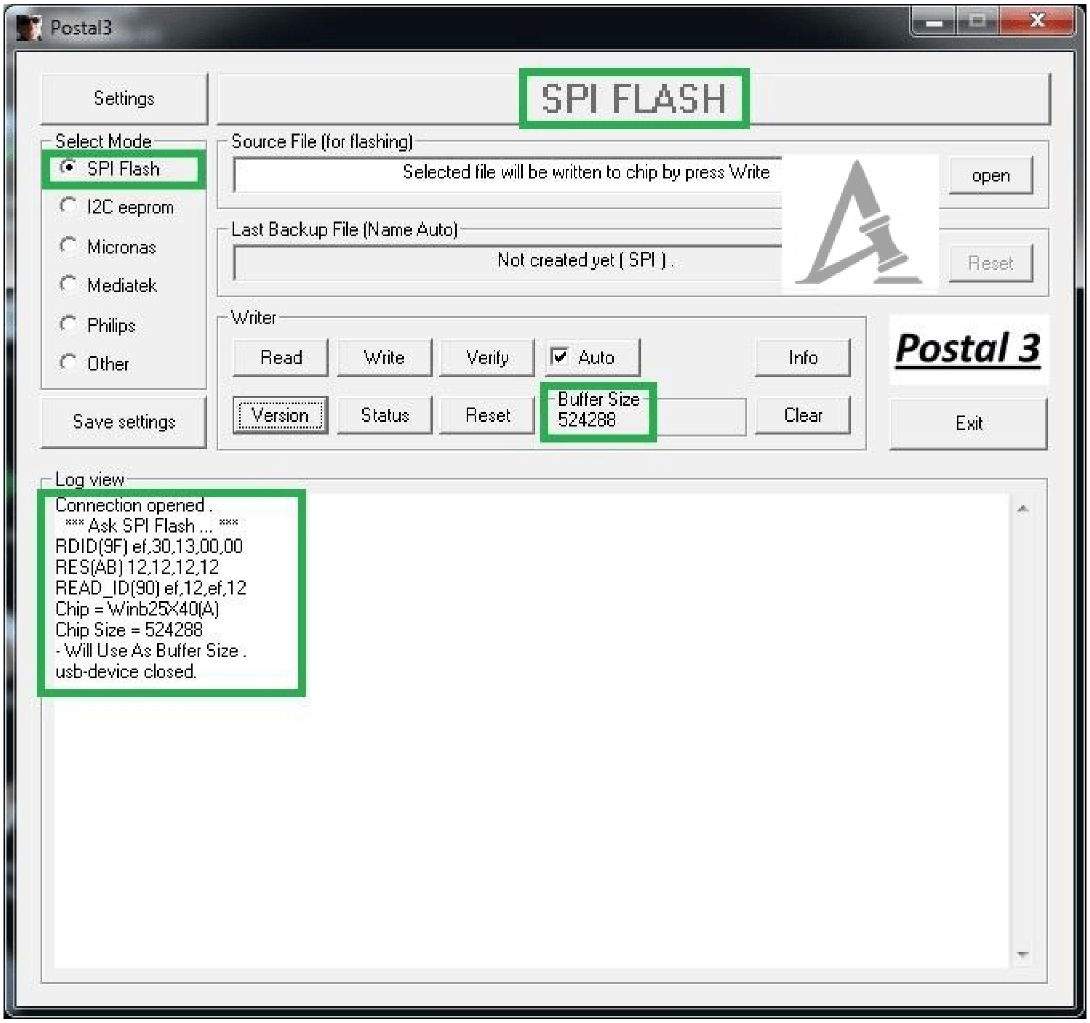
USB Driver Configuration for Running the Programmer

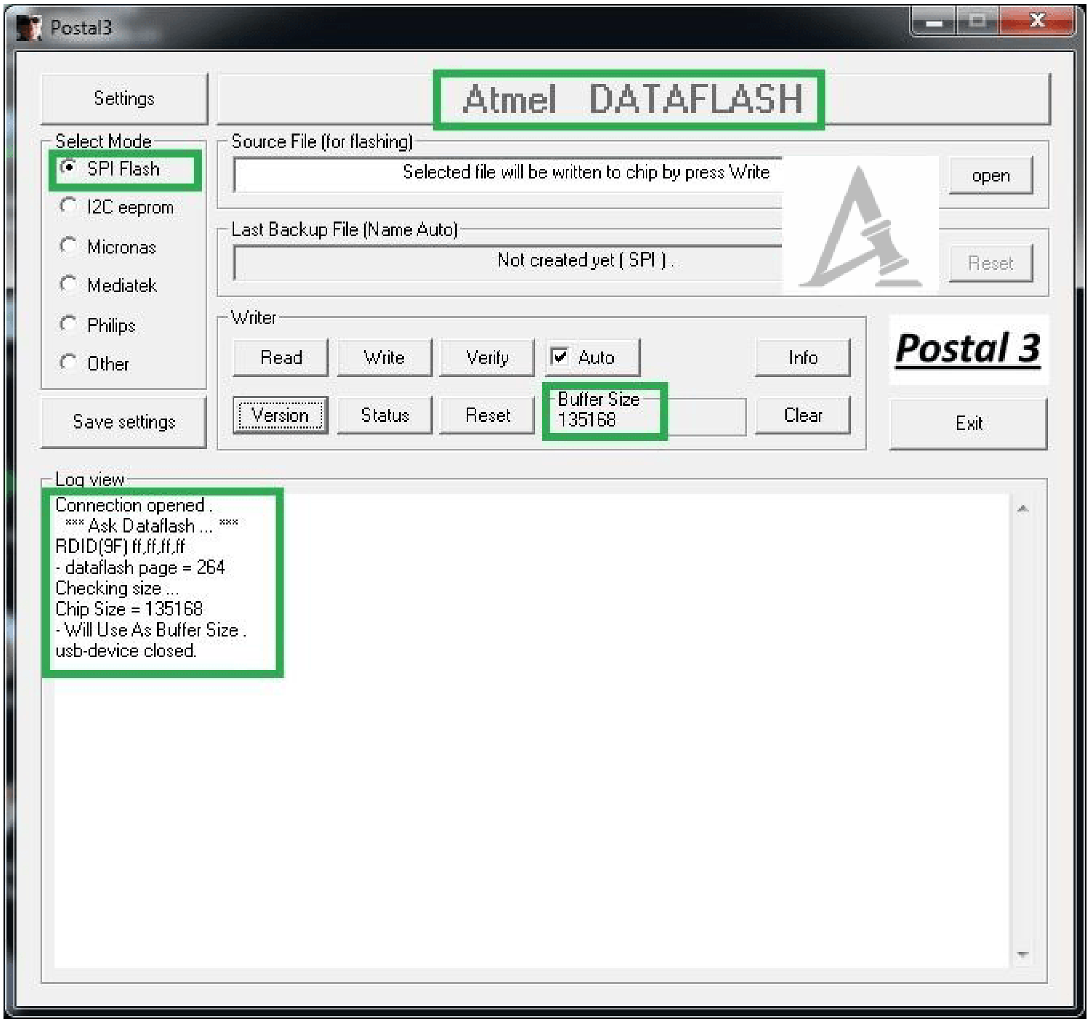


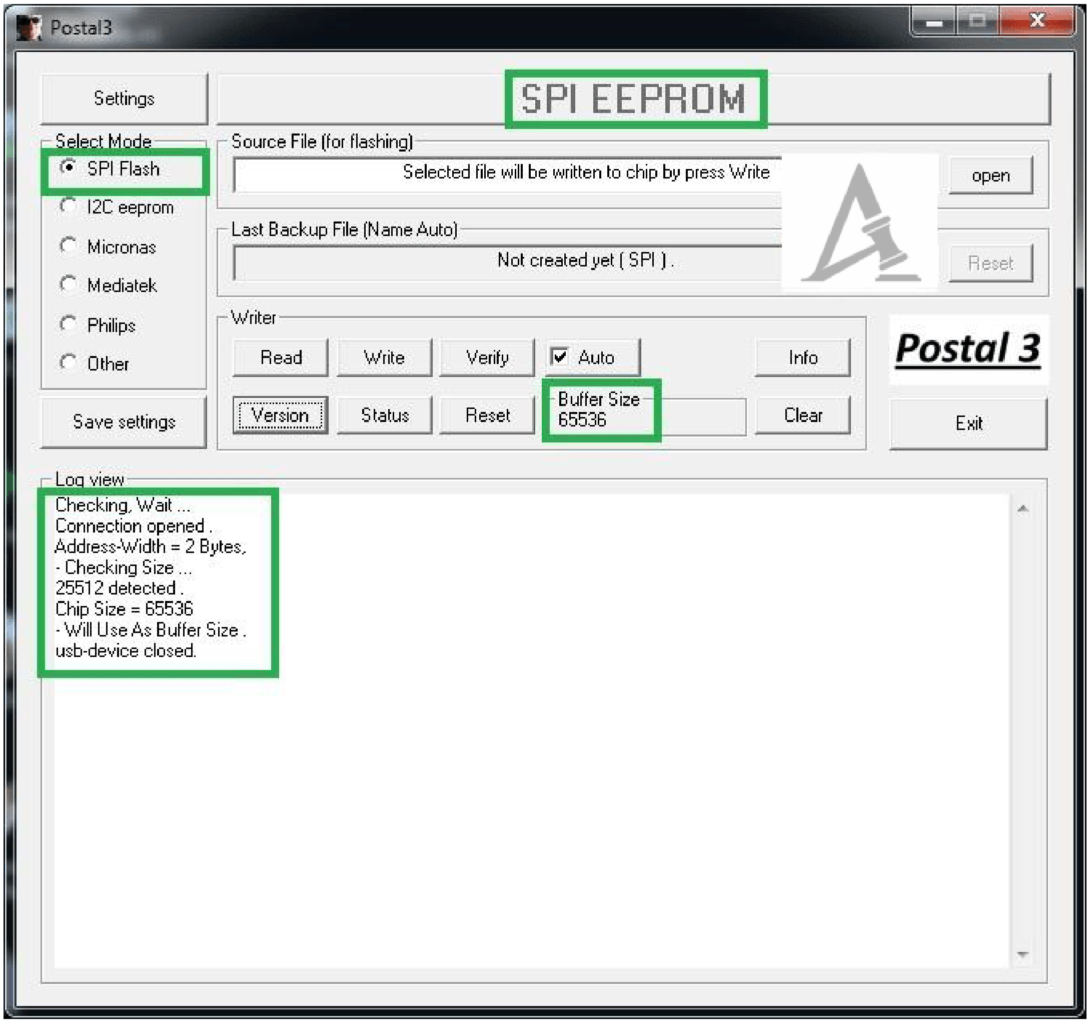
Verifying Successful Driver Installation and Software Detection of the Programmer

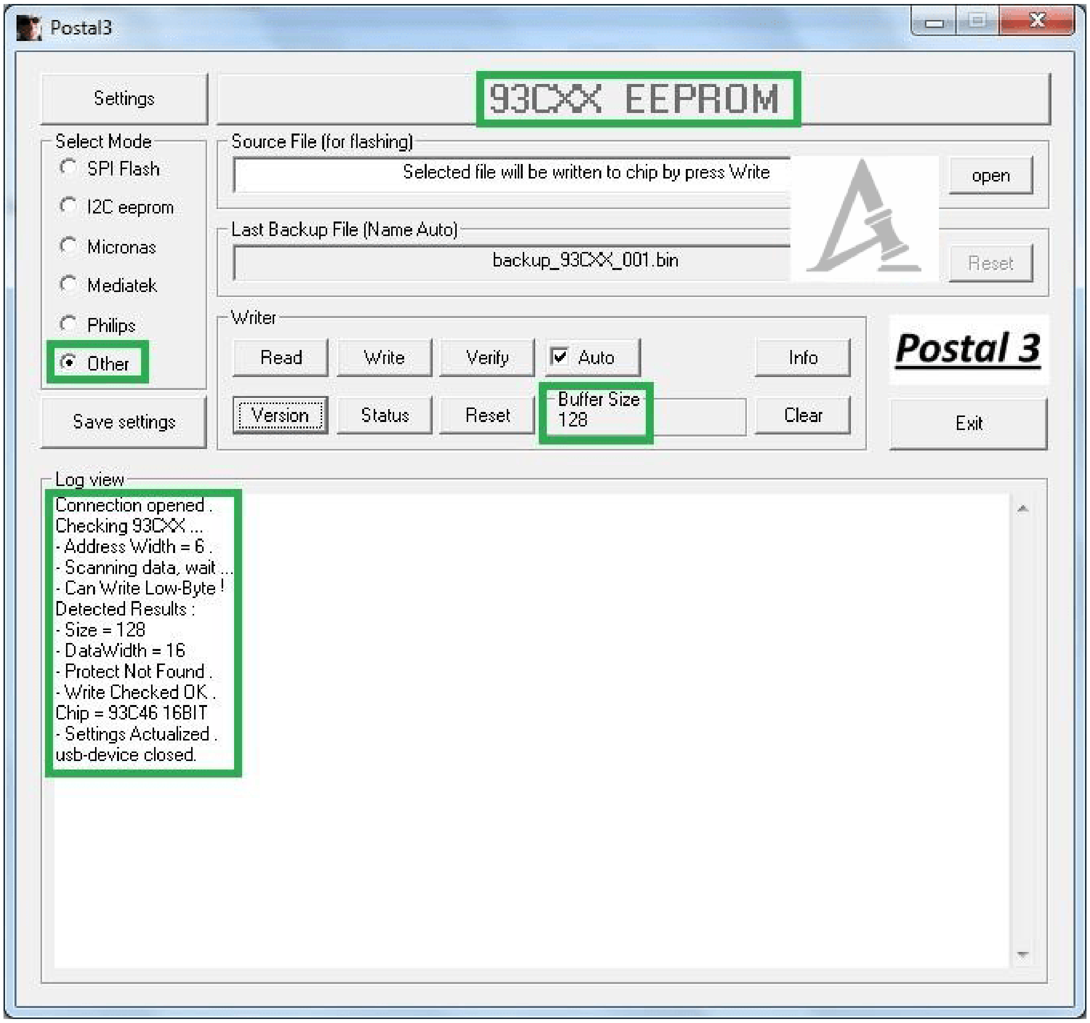
Flash and EEPROM Chips Tested in Practical Use

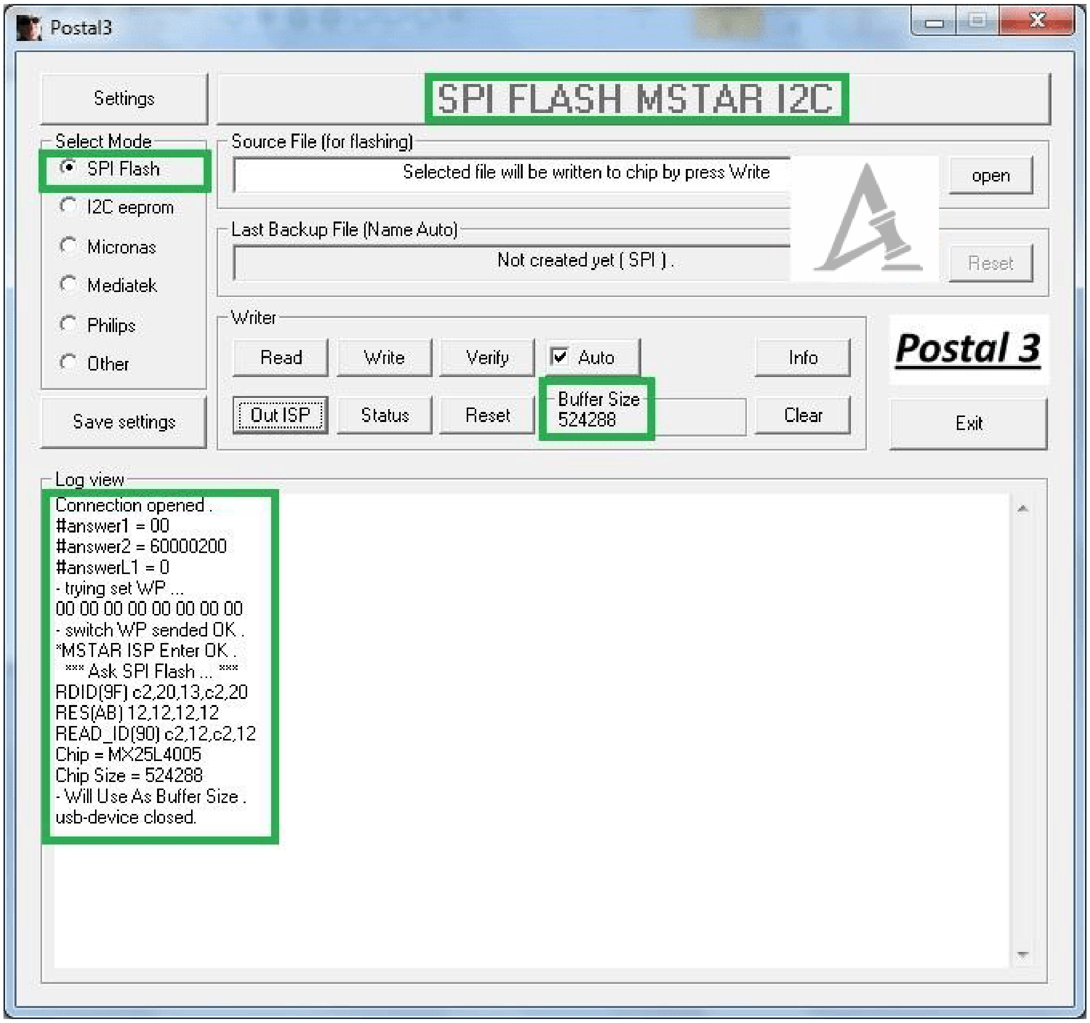

Postal3
Settings
I2C EEPROM
Select Mode
SPI Flash
I2C eeprom
Micronas
Mediatek
Philips
Other
Save settings
Source File (for flashing)
Selected file will be written to chip by press Write
open
Last Backup File (Name Auto)
Not created yet (eeprom) .
Reset
Writer
Read
Write
Verify
Auto
Info
Version
Status
Reset
Buffer Size
2048
Clear
Postal 3
Exit
Log view
Checking, Wait ...
Connection opened .
8-bit detected ...
Checking Size ...
24C16 detected .
Chip Size = 2048
- Will Use As Buffer Size .
usb-device closed.


Postal3
Settings
SPI FLASH
Select Mode
SPI Flash
I2C eeprom
Micronas
Mediatek
Philips
Other
Save settings
Source File (for flashing)
Selected file will be written to chip by press Write
open
Last Backup File (Name Auto)
Not created yet (SPI) .
Reset
Writer
Read
Write
Verify
Auto
Info
Version
Status
Reset
Buffer Size
524288
Clear
Postal 3
Exit
Log view
Connection opened .
*** Ask SPI Flash ... ***
RDID(9F) ef,30,13,00,00
RES(AB) 12,12,12,12
READ_ID(90) ef,12,ef,12
Chip = Winb25X40(A)
Chip Size = 524288
- Will Use As Buffer Size .
usb-device closed.

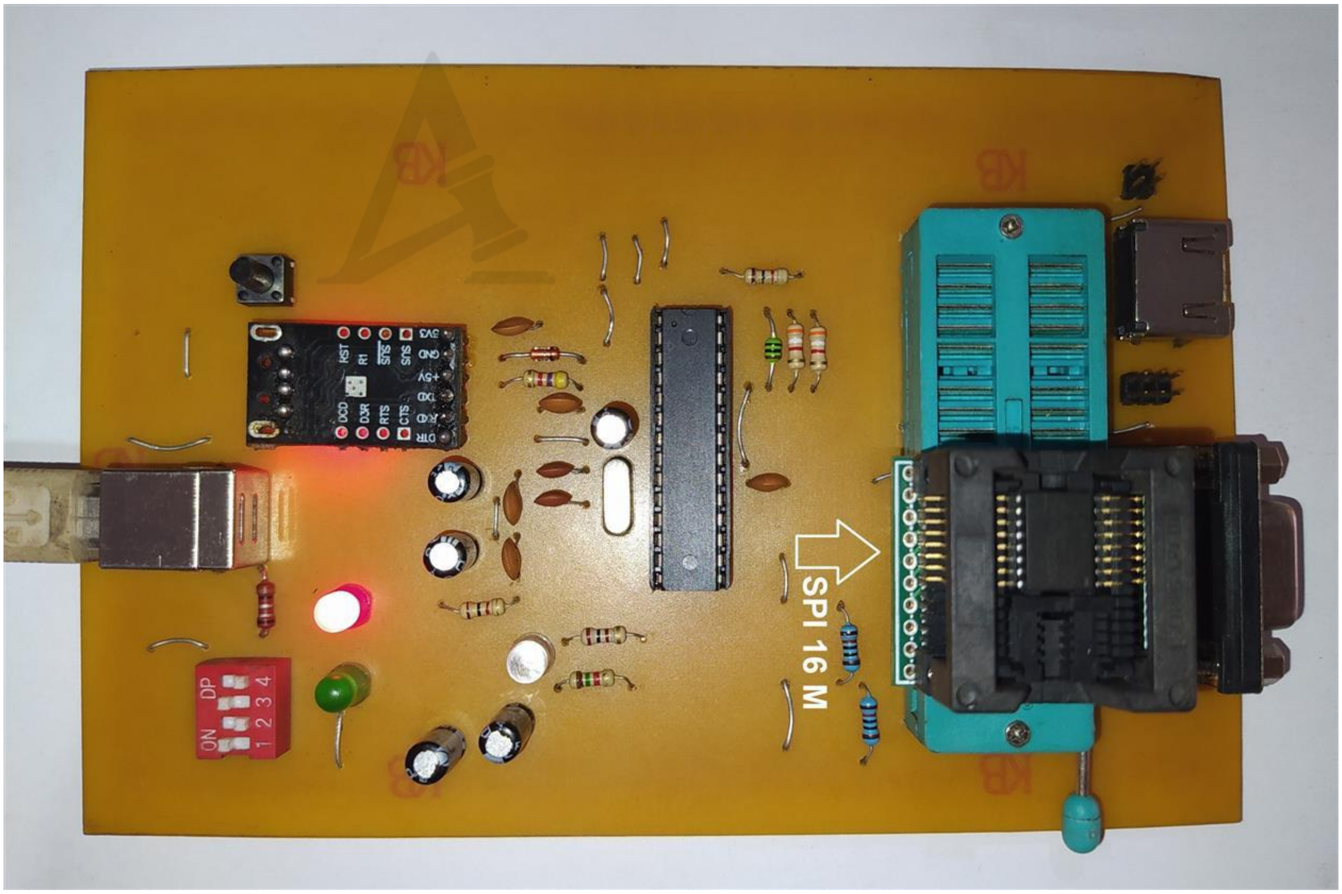

Postal3
Settings
Atmel DATAFLASH
Select Mode
SPI Flash
I2C eeprom
Micronas
Mediatek
Philips
Other
Save settings
Source File (for flashing)
Selected file will be written to chip by press Write
open
Last Backup File (Name Auto)
Not created yet (SPI) .
Reset
Writer
Read
Write
Verify
Auto
Info
Version
Status
Reset
Buffer Size
135168
Clear
Postal 3
Exit
Log view
Connection opened .
*** Ask Dataflash ... ***
RDID(9F) ff,ff,ff,ff
- dataflash page = 264
Checking size ...
Chip Size = 135168
- Will Use As Buffer Size .
usb-device closed.

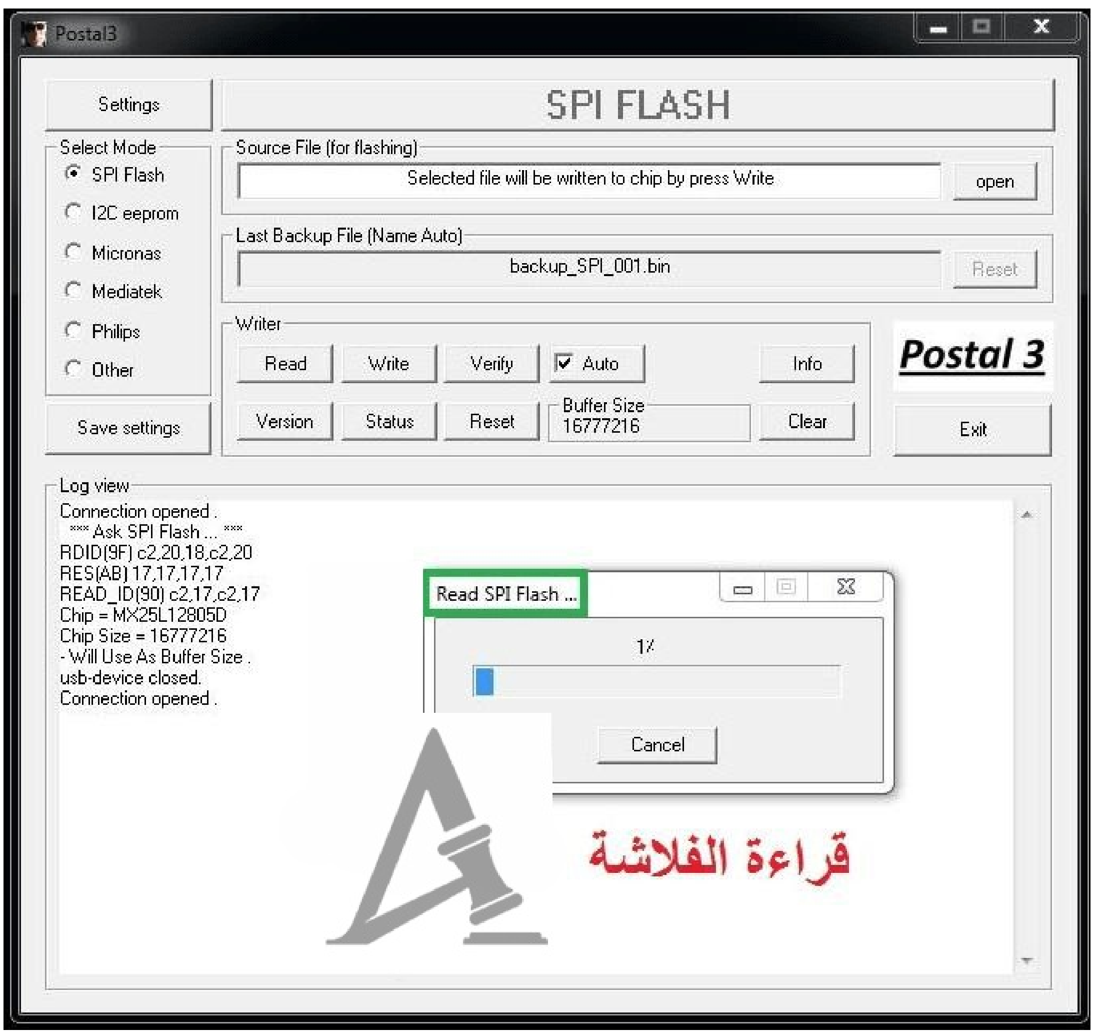

Postal3
Settings
SPI EEPROM
Select Mode
SPI Flash
I2C eeprom
Micronas
Mediatek
Philips
Other
Save settings
Source File (for flashing)
Selected file will be written to chip by press Write
open
Last Backup File (Name Auto)
Not created yet (SPI) .
Reset
Writer
Read
Write
Verify
Auto
Info
Version
Status
Reset
Buffer Size
65536
Clear
Postal 3
Exit
Log view
Checking, Wait ...
Connection opened .
Address-Width = 2 Bytes,
- Checking Size ...
25512 detected .
Chip Size = 65536
- Will Use As Buffer Size .
usb-device closed.


Postal3
Settings
93CXX EEPROM
Select Mode
SPI Flash
I2C eeprom
Micronas
Mediatek
Philips
Other
Save settings
Source File (for flashing)
Selected file will be written to chip by press Write
open
Last Backup File (Name Auto)
backup_93CXX_001.bin
Reset
Writer
Read
Write
Verify
Auto
Info
Version
Status
Reset
Buffer Size
128
Clear
Postal 3
Exit
Log view
Connection opened .
Checking 93CXX ...
- Address Width = 6 .
- Scanning data, wait ...
- Can Write Low-Byte !
Detected Results :
- Size = 128
- DataWidth = 16
- Protect Not Found .
- Write Checked OK .
Chip = 93C46 16BIT
- Settings Actualized .
usb-device closed.

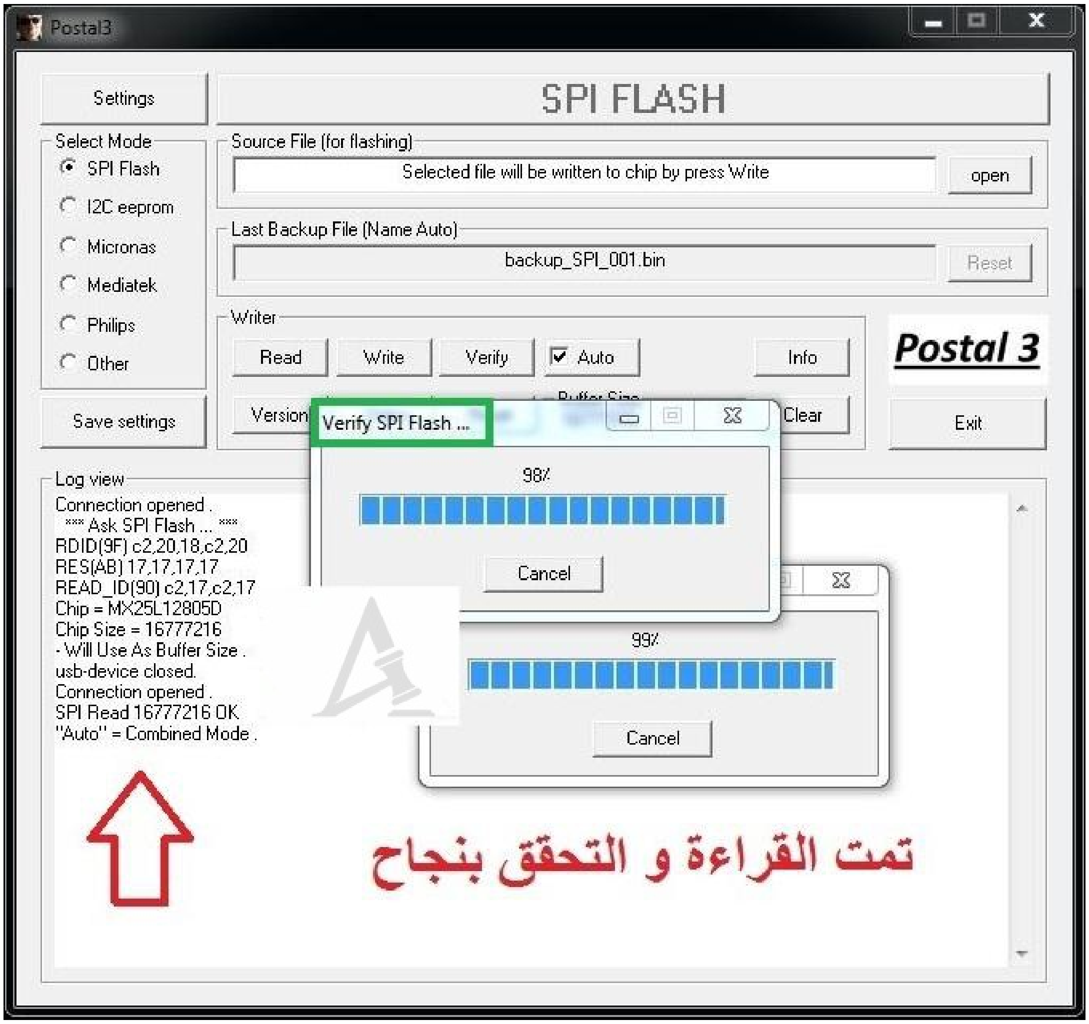


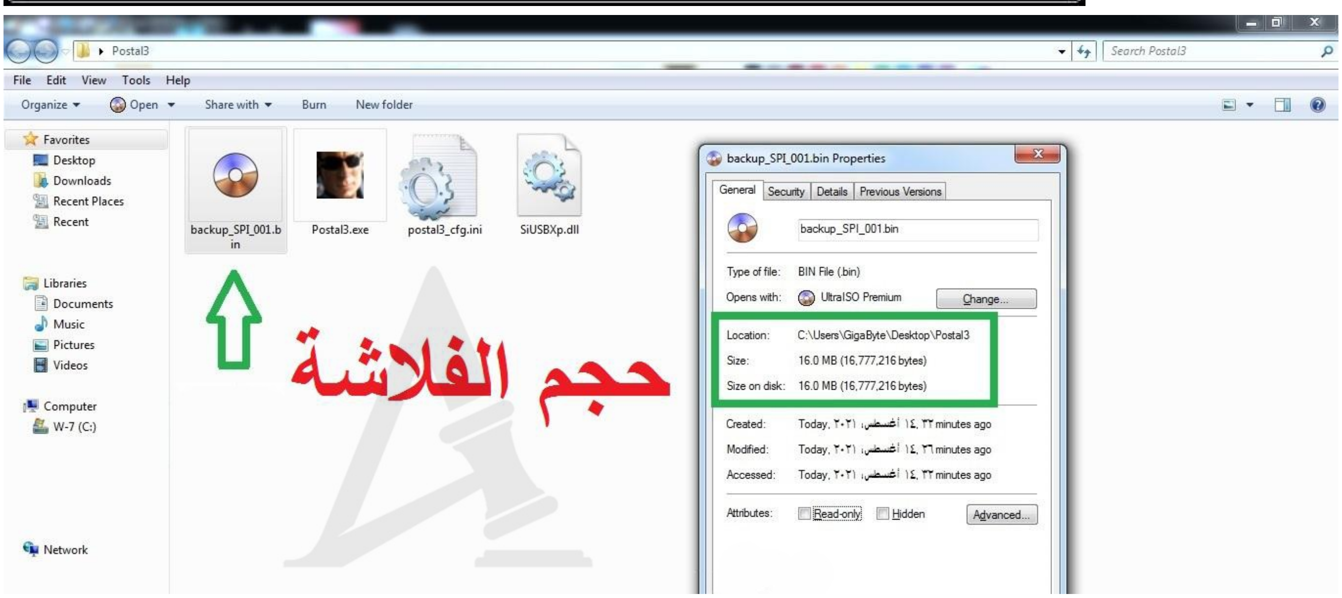
Reading Test from a 16MB SPI Flash Chip


SPI 16 M


Postal3
Settings
SPI FLASH
Select Mode
SPI Flash
I2C eeprom
Micronas
Mediatek
Philips
Other
Save settings
Source File (for flashing)
Selected file will be written to chip by press Write
open
Last Backup File (Name Auto)
backup_SPI_001.bin
Reset
Writer
Read
Write
Verify
Auto
Info
Version
Status
Reset
Buffer Size
16777216
Clear
Postal 3
Exit
Log view
Connection opened .
*** Ask SPI Flash ... ***
RDID(9F) c2,20,18,c2,20
RES(AB) 17,17,17,17
READ_ID(90) c2,17,c2,17
Chip = MX25L12805D
Chip Size = 16777216
- Will Use As Buffer Size .
usb-device closed.
Connection opened .
Read SPI Flash ...
1%
Cancel
قراءة الفلاشة


Postal3
Settings
SPI FLASH
Select Mode
SPI Flash
I2C eeprom
Micronas
Mediatek
Philips
Other
Source File (for flashing)
Selected file will be written to chip by press Write
open
Last Backup File (Name Auto)
backup_SPI_001.bin
Reset
Writer
Read
Write
Verify
Auto
Info
Postal 3
Save settings
Version
Verify SPI Flash ...
Clear
Exit
Log view
Connection opened .
*** Ask SPI Flash ... ***
RDID(9F) c2,20,18,c2,20
RES(AB) 17,17,17,17
READ_ID(90) c2,17,c2,17
Chip = MX25L12805D
Chip Size = 16777216
- Will Use As Buffer Size .
usb-device closed.
Connection opened .
SPI Read 16777216 OK
"Auto" = Combined Mode .
0%
Cancel
99%
Cancel
التحقق من الفلاشة

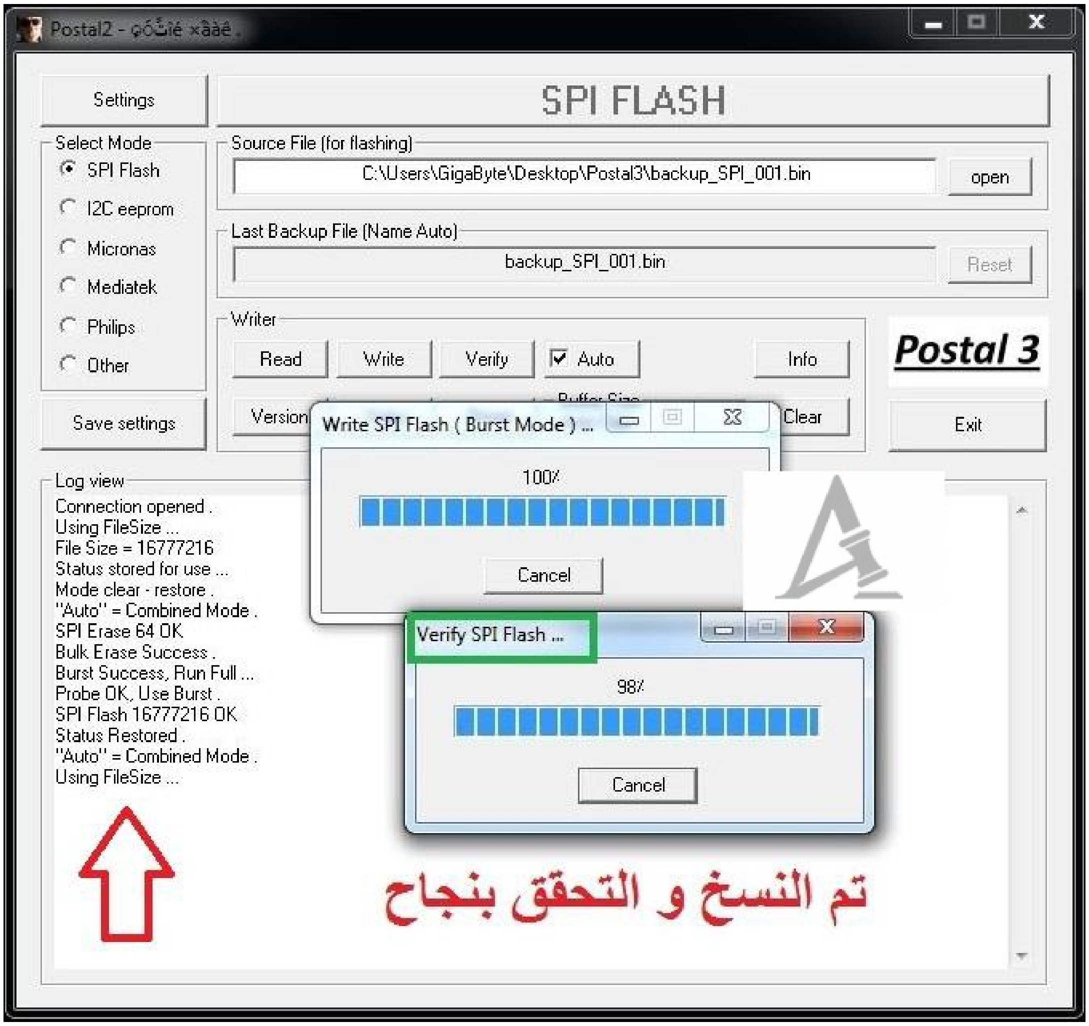




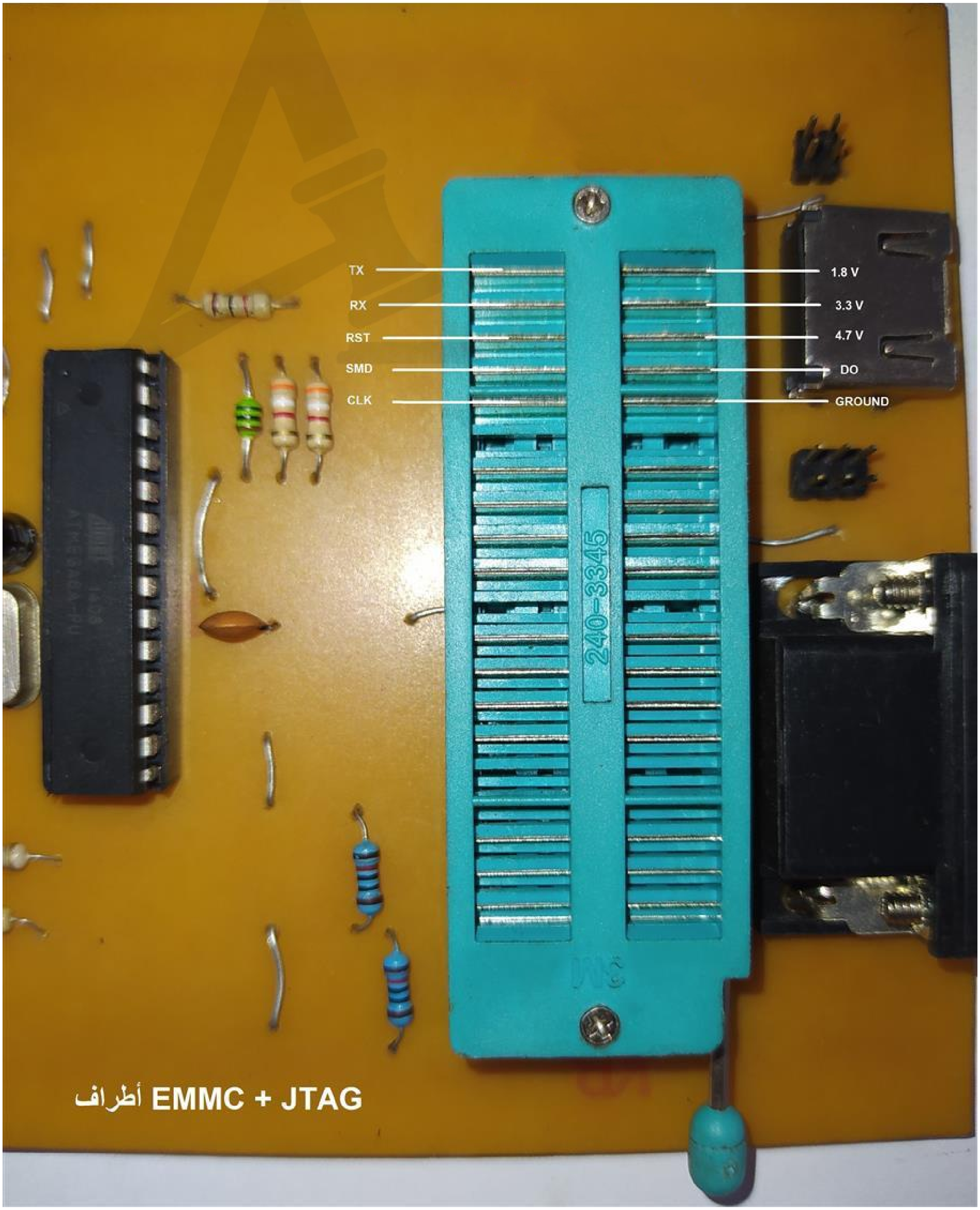
Writing Test to a 16MB SPI Flash Chip


Postal2 - ÞÓÎÉ ×ÃÈÊ
Settings
SPI FLASH
Select Mode
SPI Flash
I2C eeprom
Micronas
Mediatek
Philips
Other
Save settings
Source File (for flashing)
C:\Users\GigaByte\Desktop\Postal3\backup_SPI_001.bin
open
Last Backup File (Name Auto)
backup_SPI_001.bin
Reset
Writer
Read
Write
Verify
Auto
Info
Version
Clear
Postal 3
Exit
Wait for Busy Clear...
1%
Cancel
Log view
Connection opened .
Using FileSize ...
File Size = 16777216
Status stored for use ...
Mode clear - restore .
"Auto" = Combined Mode .
فورمات الفلاشة قبل بدء النسخ


Settings
SPI FLASH
Select Mode
SPI Flash
I2C eeprom
Micronas
Mediatek
Philips
Other
Save settings
Source File (for flashing)
C:\Users\GigaByte\Desktop\Postal3\backup_SPI_001.bin
open
Last Backup File (Name Auto)
backup_SPI_001.bin
Reset
Writer
Read
Write
Verify
Auto
Info
Version
Clear
Postal 3
Exit
Write SPI Flash (Burst Mode) ...
98%
Cancel
Log view
Connection opened .
Using FileSize ...
File Size = 16777216
Status stored for use ...
Mode clear - restore .
"Auto" = Combined Mode .
SPI Erase 64 OK
Bulk Erase Success .
Burst Success, Run Full ...
Probe OK, Use Burst .
النسخ إلى الفلاشة

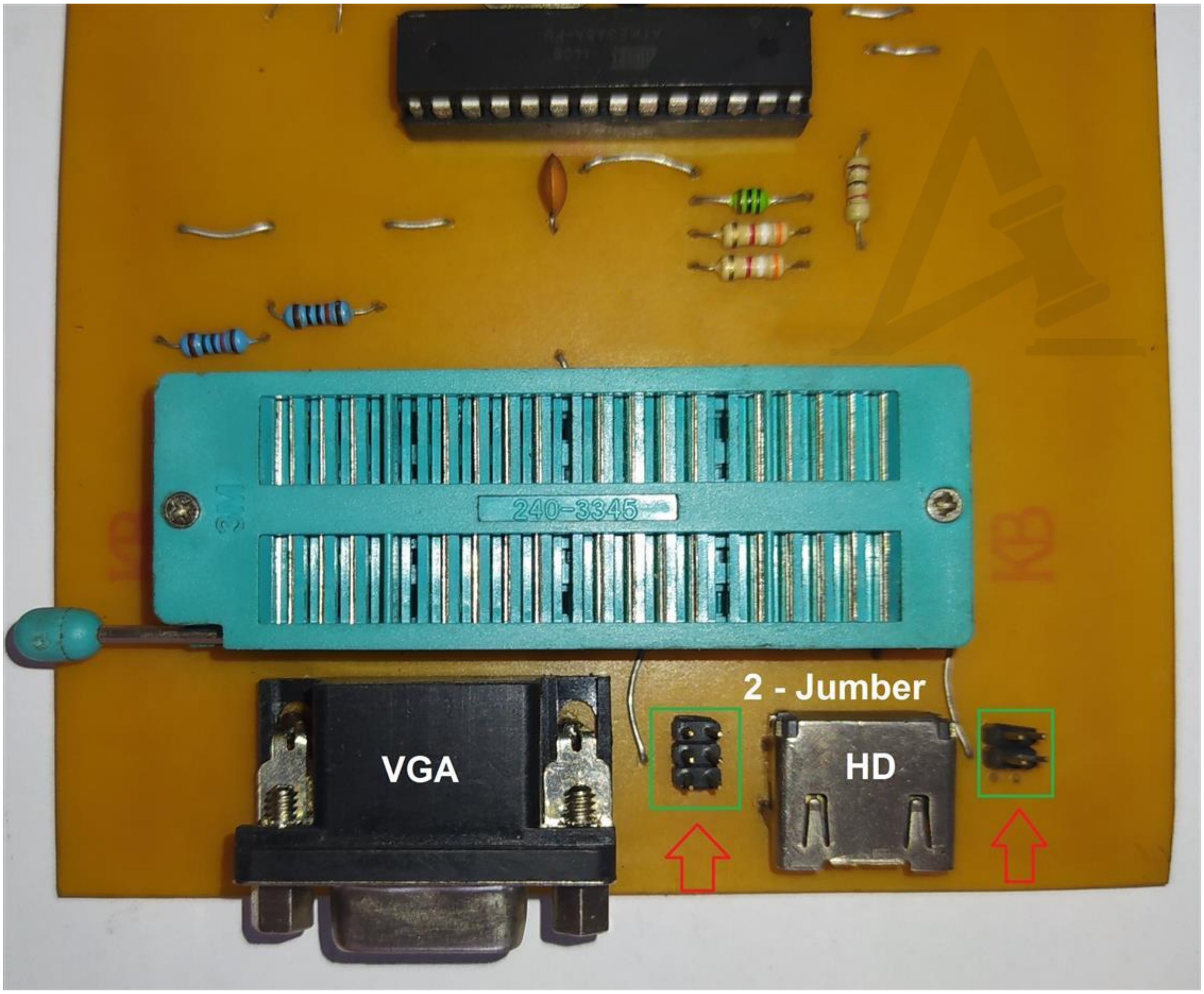


EMMC + JTAG Pinout Diagram Used for flashing TVs and receivers via board-level test points, without removing the chip from the board.



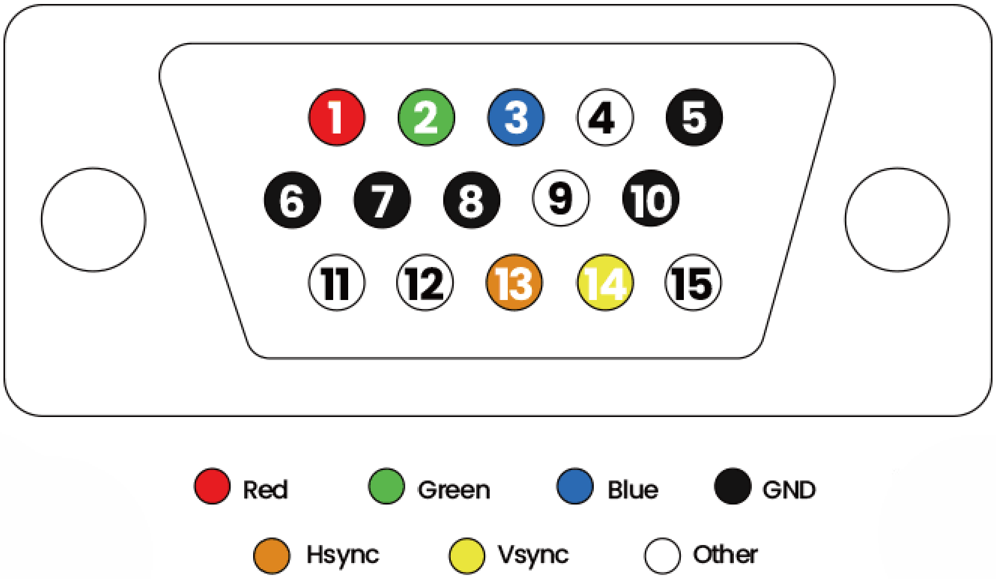
A Jumper has been added to the VGA + HDMI sockets

This allows operation across different SDA and SCL lines used in various display boards.



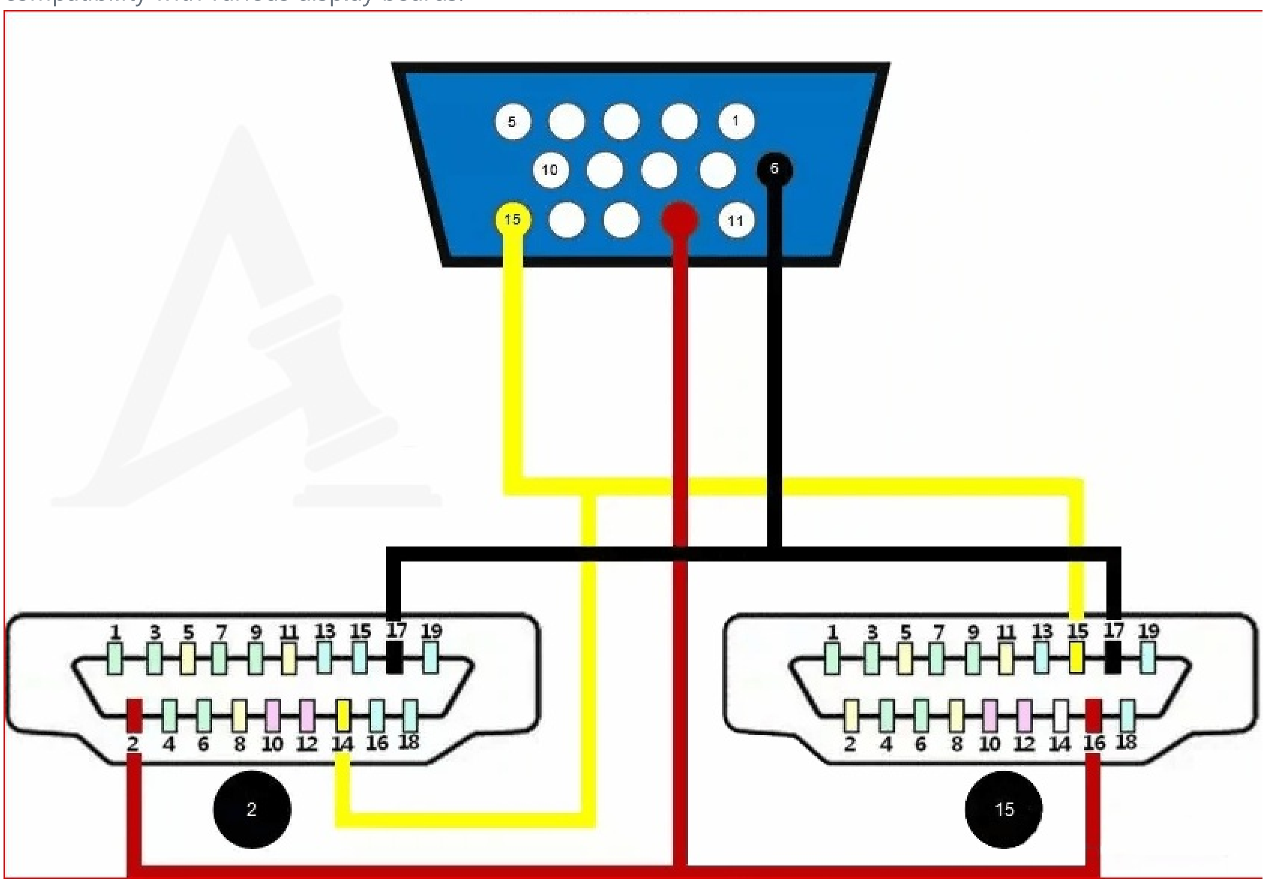
For VGA socket operation:

It enables switching between pins **4-11-5** and **12-15-5**, including reverse configurations, to ensure compatibility with different display boards.



For HDMI socket operation:

It enables switching between pins **16-15-17** and **2-14-17**, including reverse configurations, to ensure compatibility with various display boards.



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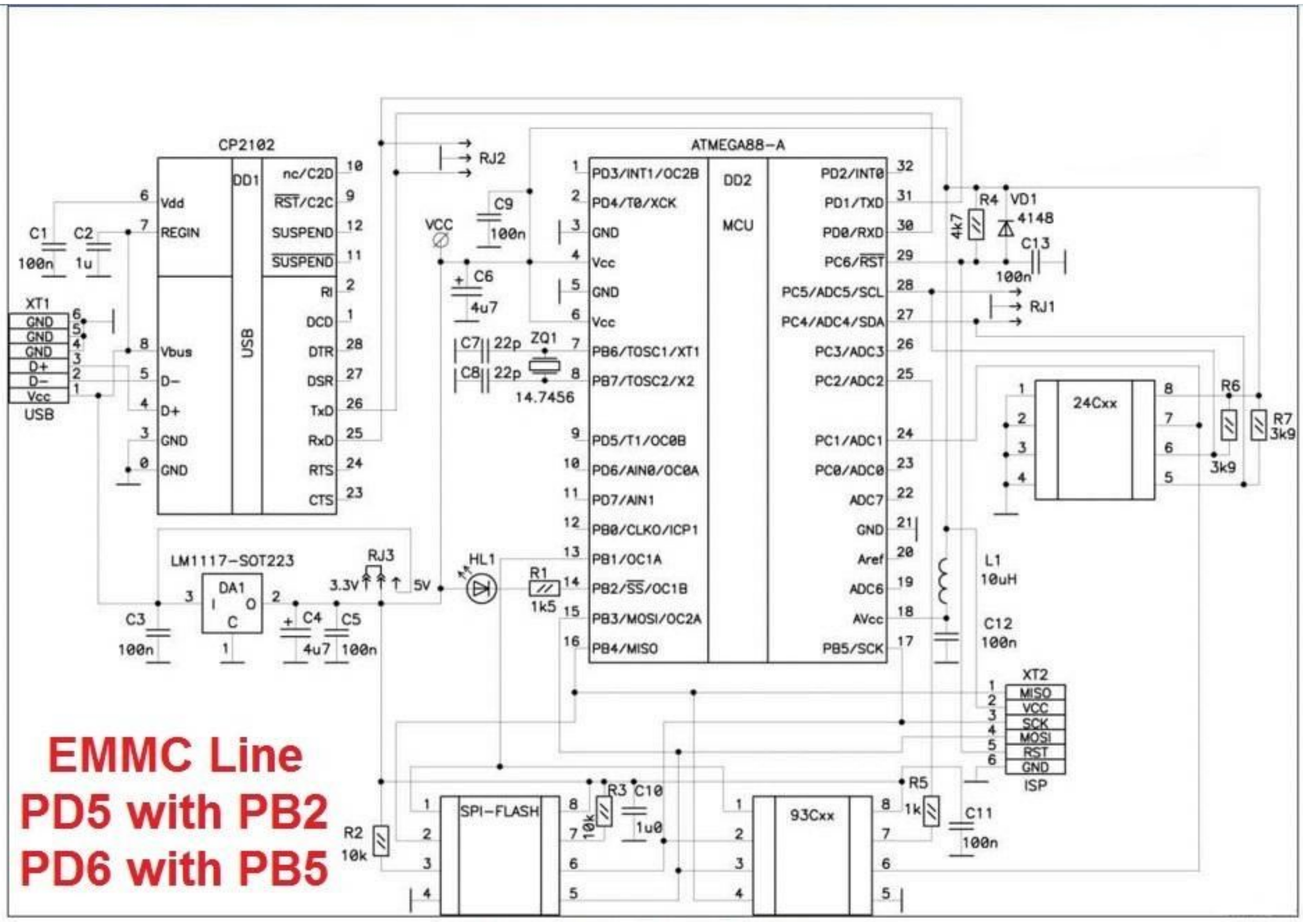
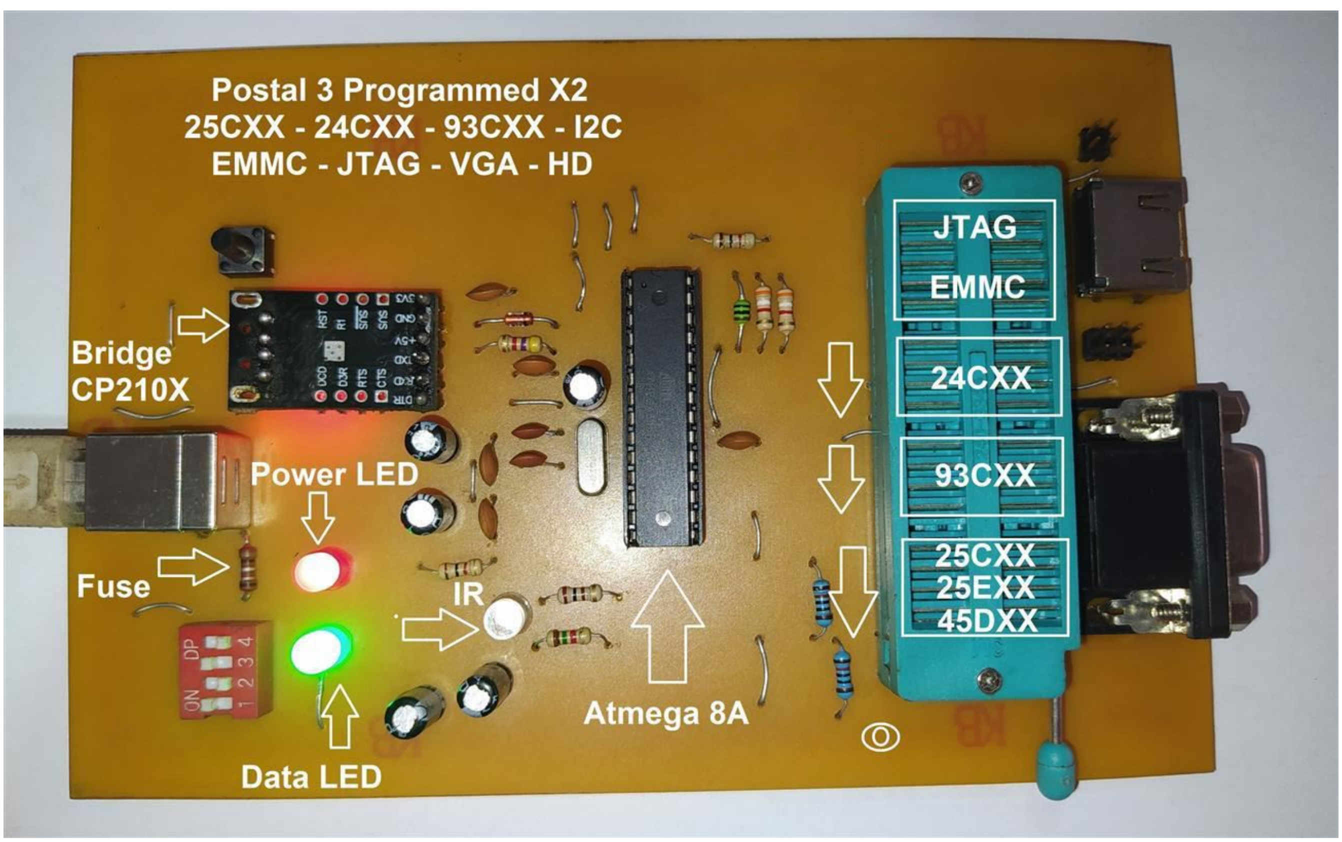
Schematic Diagram



- ==



- ===

> **Included files:**
>
> - Latest USB driver for the programmer
> - Latest POSTAL3 software version for the programmer